

OVERVIEW

Translation initiation by cap-dependent ribosome recruitment: Recent insights and open questions

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Gene expression universally relies on protein synthesis, where ribosomes recognize and decode the messenger RNA template by cycling through translation initiation, elongation, and termination phases. All aspects of translation have been studied for decades using the tools of biochemistry and molecular biology available at the time. Here, we focus on the mechanism of translation initiation in eukaryotes, which is remarkably more complex than prokaryotic initiation and is the target of multiple types of regulatory intervention. The “consensus” model, featuring cap-dependent ribosome entry and scanning of mRNA leader sequences, represents the predominantly utilized initiation pathway across eukaryotes, although several variations of the model and alternative initiation mechanisms are also known. Recent advances in structural biology techniques have enabled remarkable molecular-level insights into the functional states of eukaryotic ribosomes, including a range of ribosomal complexes with different combinations of translation initiation factors that are thought to represent *bona fide* intermediates of the initiation process. Similarly, high-throughput sequencing-based ribosome profiling or “footprinting” approaches have allowed much progress in understanding the elongation phase of translation, and variants of them are beginning to reveal the remaining mysteries of initiation, as well as aspects of translation termination and ribosomal recycling. A current view on the eukaryotic initiation mechanism is presented here with an emphasis on how recent structural and footprinting results underpin axioms of the consensus model. Along the way, we further outline some contested mechanistic issues and major open questions still to be addressed.

This article is categorized under:

Translation > Translation Mechanisms

Translation > Translation Regulation

RNA Interactions with Proteins and Other Molecules > Protein–RNA

Interactions: Functional Implications

KEYWORDS

eukaryotic translation initiation, mRNA translation, post-transcriptional gene regulation, protein synthesis, ribosome, scanning, start codon recognition

1 | INTRODUCTION

Molecular biology as a discipline was formed in the 1950s around the “central dogma”, and its foundational studies were directed at understanding the processes of transcription of genetic information from DNA into messenger (m)RNA and translation from mRNA into protein. Understanding the genetic code and the roles of transfer (t)RNA and the ribosome in decoding it during protein synthesis were key early advances. It was also realized that ribosomes consist of small and large

subunits (SSU and LSU, respectively). Most of the decoding-related functions of the ribosome involve regions of the SSU, including the tRNA-binding sites for the incoming aminoacyl-tRNA (A-site), the peptidyl-tRNA (P-site), the outbound deacylated tRNA (E-site), and the mRNA interface (Korostelev & Noller, 2007; Steitz & Moore, 2017; Wilson & Doudna Cate, 2012). Conversely, the LSU contains the peptidyl transferase center responsible for the polymerization of amino acids, as well as surfaces required for handling the nascent polypeptide (Steitz, 2005). There also was an early realization that the recruitment of the ribosome to the mRNA was an important “initiation” phase during translation, with mechanistic requirements that were distinct from “elongation”, the actual protein synthesis phase, as well as “termination”, where ribosomes stop protein synthesis at nonsense codons and dissociate into free subunits to enable a new translation round (Figure 1a).

Many aspects of translation are well conserved throughout evolution. For example, the prokaryotic and eukaryotic elongation and termination phases feature comparable mechanisms and involve broadly similar sets of elongation factors (EFs) and release factors (RFs), respectively (Frank & Gonzalez Jr., 2010; Spirin, 2009b). Other aspects are distinctly different, such as the spatial separation of transcription and translation into the nucleus and cytoplasm of eukaryotes, respectively, while prokaryotic mRNAs are translated cotranscriptionally. Furthermore, apart from a conserved stepwise recruitment of ribosomal subunits, prokaryotic and eukaryotic initiation are mechanistically quite divergent. Notably, while three initiation factors (IFs) suffice for the process in prokaryotes, evolution has generated a plethora of additional eukaryotic (e)IFs (Marintchev & Wagner, 2004). Related to this, prokaryotic mRNAs are typically polycistronic and have short untranslated regions (UTRs), while eukaryotic mRNAs usually contain one main open reading frame (ORF) and possess rather longer 5' and 3' UTRs (Hinnebusch, Ivanov, & Sonenberg, 2016).

Extensive research into eukaryotic initiation commenced in the 1970s (Anderson et al., 1974). The ubiquitous presence of an “inverted” 3' → 5' 7-methylguanosine m⁷G(5')ppp(5')N “cap” structure at the 5' end of eukaryotic mRNAs was discovered (Furuichi, 2014). Caps are added to mRNAs cotranscriptionally while still near transcription start sites, by cleaving the γ-phosphate off the mRNA 5' end followed by GMP transfer from GTP by guanylyltransferase and its N⁷-methylation by (guanine-7-)methyltransferase (Furuichi, 2014). In higher eukaryotes, the first and second encoded mRNA nucleotides can then be additionally 2'-methylated in the nucleus and cytoplasm, respectively (Furuichi, 2014). Caps can also be added post-

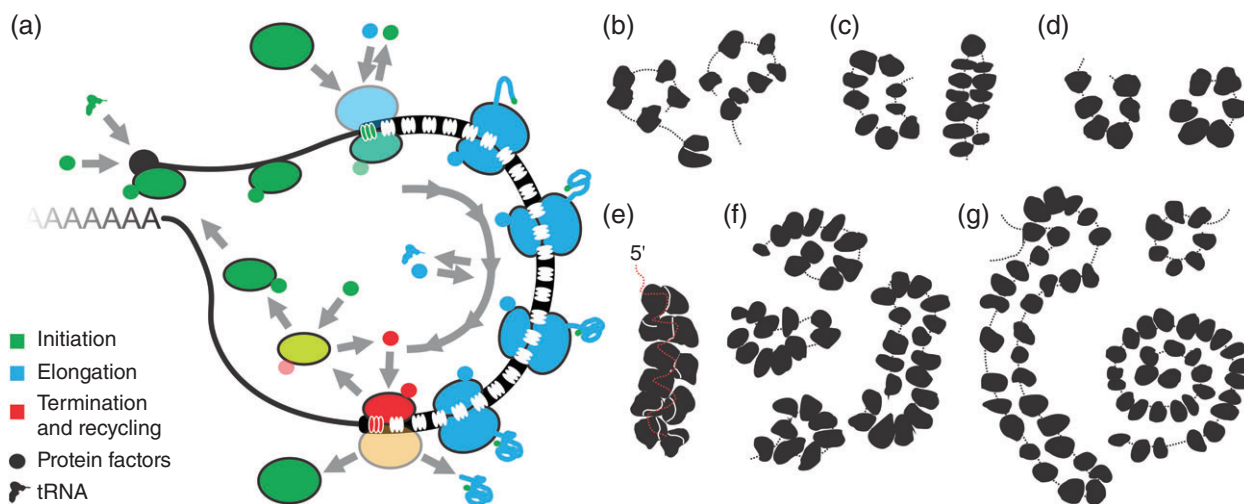


FIGURE 1 (a) Schematic of the eukaryotic translation cycle showing a single mRNA translated by multiple ribosomes forming a polysome. Canonical translation initiation (green) begins with the assembly of the small ribosomal subunit (SSU), eukaryotic initiation factors (eIFs), and initiator methionyl-tRNA (Met-tRNA_i^{Met}) at the mRNA 5' end, assisted by 5' cap to 3' poly(A) interactions. The SSU complex then moves in a 3' direction until it recognizes a start codon, rearranges its configuration, and is joined by the large ribosomal subunit (LSU). This commences elongation (blue), whereby ribosomes move through the open reading frame (ORF) and synthesize the polypeptide, aided by elongation factors (eEFs). Termination (red) occurs at nonsense (stop) codons; here, eukaryotic release factors (eRFs) assist in the release of the completed protein as well as the LSU and SSU, which can be recycled for additional rounds of translation. The gradual displacement of eIFs at start codons is depicted in light moss green (●). Because SSUs at start codons can belong to both initiation and elongation complexes, they are shown in amazonite (●). The sequential nature of ribosome assembly during initiation (LSU attaches last) and disassembly during termination (LSU leaves first) is shown by sky blue (●) and caramel (●) colors, respectively. Sequential displacement of eRFs with initially recycling followed by initiation factors upon completion of termination is depicted by citron (●) SSU and pink-colored (●) eRFs. (b)–(g) Schematics of polysome topologies as detected by electron microscopy. (b) “circular” polysomes observed in rabbit reticulocyte lysate on native mRNA with 5' cap and 3' poly(A) (Warner, Rich, & Hall, 1962). (c) “circular” and “hairpin” polysomes seen in *Tetrahymena pyriformis* cells (Yazaki, Yoshida, Wakiyama, & Miura, 2000). (d) “circular” polysomes formed in a wheat germ cell-free translation system on capped and polyadenylated (left) and noncapped, nonpolyadenylated (right) synthetic mRNAs (Afonina, Myasnikov, Shirokov, Klaholz, & Spirin, 2014). (e) “linear” polysome formed in a wheat germ cell-free translation system on capped but nonpolyadenylated mRNA, featuring interacting SSUs along the central axis of a left-handed (5' to 3') spiral of mRNA (Myasnikov et al., 2014). (f) Endoplasmic reticulum (ER)-attached “circular”, “linear”, “double-row”, and “spiral” polysomes detected in rat hepatocytes (Palade, 1975). (g) ER-attached “circular”, “hairpin”, and “spiral” polysomes from rat fibroblasts showing SSU-inward orientation of ribosomes in the tightly packed curved regions (Christensen & Bourne, 1999)

transcriptionally by different eukaryotic viruses (Decroly & Canard, 2017). Roles of the cap structure in mRNA stability (Shimotohno, Kodama, Hashimoto, & Miura, 1977) and translation (Muthukrishnan, Both, Furuichi, & Shatkin, 1975; Shafritz et al., 1976) were realized soon after its discovery; it is now widely accepted that uncapped mRNA in eukaryotic cytoplasm rapidly degrades (Arribas-Layton, Wu, Lykke-Andersen, & Song, 2013; Labno, Tomecki, & Dziembowski, 2016). However, early mechanistic explanations for the attachment of SSUs to mRNA during eukaryotic initiation lacked the elegance of the model established in parallel for prokaryotes, where a specific sequence (AGGAGGU in *Escherichia coli*) is found in mRNA just upstream of start codons. This Shine–Dalgarno sequence was immediately proposed (Shine & Dalgarno, 1974, 1975) and later structurally demonstrated (Kaminishi et al., 2007; Yusupova, Yusupov, Cate, & Noller, 2001) to recruit the SSU to mRNA through Watson–Crick interactions with the 3′ end of the 16S ribosomal (r)RNA. Initially, it seemed that the only determinant of the start site in eukaryotes was the AUG codon itself, to which ribosomes can bind directly (Baralle & Brownlee, 1978). The overall process of initiation seemed to be less organized than in bacteria (Heywood, 1973).

With rare exceptions, such as metazoan replication-dependent histone mRNAs, eukaryotic mRNAs also carry a 3′ poly(A) “tail”, which is added cotranscriptionally in a two-step process. Nascent pre-mRNA is first cleaved after recognition of the poly(A) site (AAUAAA, located 10–30 nt upstream of the cleavage site), auxiliary signals (CA sequence before and U/GU-rich element ~40 nt after the cleavage site), and U-rich elements with UGUA consensus (upstream the cleavage site) (Shi & Manley, 2015). Recognition and processing is aided by interactions with the RNA polymerase II C-terminal domain and involve cleavage and polyadenylation specificity factor, cleavage stimulation factor, cleavage factors I and II, symplekin, and poly(A) polymerase (Shi & Manley, 2015). Poly(A) is then added from ATP to ~250 nt in length (Viphakone, Voisinnet-Hakil, & Minvielle-Sebastia, 2008). mRNAs with longer 3′ poly(A) are usually better translated, whereas a shorter 3′ poly(A) often results in decreased translation and removal of the mRNA from the translated pool (Beilharz & Preiss, 2007; Richter, 2007; Villalba, Coll, & Gebauer, 2011). Notably, alternative polyadenylation sites can be preferentially selected under different conditions; they can define additional 3′ UTR complexity and include or exclude RNA-binding protein (RBP) sites, microRNA-binding sites, and even exons and introns (Elkon, Ugalde, & Agami, 2013; Soetanto et al., 2016; Turner, Pattison, & Beilharz, 2017). The 3′ poly(A) length can further be dynamically regulated in the cytoplasm, a process highly important for post-transcriptional gene control (Charlesworth, Meijer, & de Moor, 2013; D’Ambrogio, Nagaoka, & Richter, 2013; Jalkanen, Coleman, & Wilusz, 2014; Weill, Belloc, Bava, & Mendez, 2012).

Importantly, the 5′ cap and 3′ poly(A) allow the formation of an mRNA “closed-loop”, facilitated by interactions between the cap-binding eIFs and the poly(A)-binding protein (PABP) (Archer, Shirokikh, Hallwirth, Beilharz, & Preiss, 2015; Preiss & Hentze, 1999; Tarun Jr. & Sachs, 1996). The closed-loop promotes recruitment of free SSUs from the cytosol and may also be involved in the direct recycling of SSUs from termination to initiation within established polysomes. Polysomes are “steady-state” assemblies of mRNAs translated by serially elongating ribosomes. The spatial configurations seen for polysomes from different sources are mostly consistent with the closed-loop model, although some of the observed shapes suggest additional or alternative intrapolysomal interactions (Figure 1b–g).

Decades of biochemistry and molecular biology research have shed much light on the translation process, revealing numerous mechanistic details. However, key contributions have increasingly been made by the structural biology field. Solving the crystal structure of the ribosome was a key advance (Ban, Nissen, Hansen, Moore, & Steitz, 2000; Cate, Yusupov, Yusupova, Earnest, & Noller, 1999; Clemons Jr. et al., 1999; Tocilj et al., 1999), and the structures of multiple auxiliary factors (eIF, eEFs, eRFs) have provided much useful information. The advent of cryo-electron microscopy (cryo-EM) approaches then facilitated the detailed structural characterization of several ribosome–factor–mRNA complexes thought to represent functional intermediates of translation (Frank, 2017). In parallel, ribosome profiling, a combination of the purification of arrested ribosomal complexes from living cells with transcriptome-wide sequencing of their RNase-protected “footprints” on mRNA, provided much complementary insight (Archer, Shirokikh, Beilharz, & Preiss, 2016; Ingolia, Ghaemmaghani, Newman, & Weissman, 2009; McGlincy & Ingolia, 2017). In this review, we focus on what is known about the mechanism of canonical cap-dependent, scanning-driven translation initiation, drawing primarily on plentiful structural and biochemical data available for mammalian and yeast cells. We outline key discoveries and concepts and detail how their validity is underpinned by recent cryo-EM, crystallographic, and ribosome footprinting data. We emphasize areas of agreement and discord between new and old data, as well as aspects of mechanistic uncertainty that call for further investigation.

2 | OVERVIEW OF CAP-DEPENDENT TRANSLATION INITIATION AND ITS REGULATION

Combining the scanning model with the accumulated knowledge of the identities, interactions, and other biochemical activities of the eIFs allowed the construction of what might be a predominant sequence of events during initiation (see Table 1 for the list of main protein factors directly involved in the ribosome–mRNA interactions throughout translation initiation). An overview of this “canonical” pathway is depicted in Figure 2. To better represent the topological aspects of the model,

TABLE 1 List of eukaryotic translation initiation factors participating in the consensus initiation pathway, as well as some other proteins with roles in initiation^a

Initiation factor	Name in yeast	Also known as	MW, kDa (human) ^b	Key function in initiation	References
eIF1	<i>SUI1</i> / YNL244C	eIF-1, IF-E1, A121, EIF1A, ISO1	12.7	Destabilization of codon-anticodon interactions	Asano (2014), Fijalkowska et al. (2017), Hinnebusch (2017), Jackson, Hellen, and Pestova (2010), Lomakin and Steitz (2013), Obayashi et al. (2017)
eIF1A	<i>TIF11</i> / YMR260C	eIF-1A, EIF1AX, EIF1A, EIF1API, eIF-4C, EIF4C, IF-E7, IF-M2B β	16.5	Control of P _N /P _{OUT} switching during scanning and start codon recognition	Asano (2014), Hinnebusch (2017), Jackson et al. (2010), Lind and Aqvist (2016), Lomakin and Steitz (2013), Nag et al. (2016)
eIF2 (composite)					
		eIF-2, IF-MP, IF-1, IF-E2, EIF-3	125.6	Positioning of Met-tRNA _i ^{Met} in the SSU P-site during scanning, recognition of the AUG and near-cognate start codons (GUG, AUA, etc.), and the consensus nucleotide context	Asano (2014), Hinnebusch (2017), Jackson et al. (2010), Sokabe and Fraser (2014)
eIF2 α	<i>SUI2</i> / YJR007W	EIF2S1, EIF-2, EIF-2A, EIF2alpha, EIF2, EIF2A	36.1	eIF2 availability by Ser 51 phosphorylation and increased binding to the GDP/GTP exchange factor eIF2B; recognition of the downstream start codon context, and stabilization of P _N	Jennings, Kershaw, Adomavicius, and Pavitt (2017), Visweswarajah and Hinnebusch (2017)
eIF2 β	<i>SUI3</i> / YPL237W	EIF2S2, EIF2, EIF2B, EIF2beta, PPP1R67	38.4	Binding to Met-tRNA _i ^{Met} and eIF5, eIF2 availability by binding to the GDP/GTP exchange factor eIF2B	Jennings et al. (2016)
eIF2 γ	<i>GCD11</i> / YER025W	EIF2S3, EIF2, EIF2G, EIF2gamma, eIF-2 γ A, MRXSBRK	51.1	GTP/GDP binding and GTP hydrolysis	Alone and Dever (2006)
eIF2A	YGR054W	eIF-2A, EIF-2A, IF-M1, MST089, MSTP004, MSTP089, CDA02	65.0	GTP-independent Met-tRNA _i ^{Met} , Leu-tRNA _i ^{Leu} binding, and start codon recognition, including noncanonical CUG and UUG triplets on specific mRNAs	Kearse and Wilusz (2017), Sendoel et al. (2017)
eIF2D	<i>TMA64</i> / YDR117C	Ligatin, HCA56, LGTN	64.7	GTP-independent Met-tRNA _i ^{Met} binding and start codon recognition	Dmitriev et al. (2010), Vaidya, Lomakin, Joseph, Dmitriev, and Steitz (2017)
eIF3 (composite)					
		IF-3, IF-E3, IF-M5, eIF-3	793.6	Attachment to the cap, assembly of the scanning complex, coordination of cap-dependent mRNA loading, scanning, start codon recognition, and LSU joining	Asano (2014), Cate (2017) Erzberger et al. (2014), Hinnebusch (2017), Jackson et al. (2010), Obayashi et al. (2017), Valasek et al. (2017)
eIF3a	<i>RPG1</i> / YBR079C	eIF3A, P167, eIF3-p170, eIF3-theta, p180, p185	166.6	MPN/PCI subunit. Interacts with mRNA	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3b	<i>PRT1</i> / YOR361C	eIF3B, EIF3-eta, EIF3-P110, EIF3-P116, EIF3S9	92.5	Peripheral subunit. Interacts with mRNA	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3c	<i>NIP1</i> / YMR309C	EIF3C, EIF3CL, EIF3S8, eIF3-p110	105.3	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3d	N/A	EIF3D, EIF3S7, eIF3-p66, eIF3-zeta	64.0	Peripheral subunit. Direct binding to the cap structures for specific mRNAs. Interacts with mRNA	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3e	N/A	EIF3E, EIF3-P48, EIF3S6, INT6, eIF3-p46	52.2	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3f	N/A	EIF3F, EIF3S5, eIF3-p47	37.6	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)

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TABLE 1 (Continued)

Initiation factor	Name in yeast	Also known as	MW, kDa (human) ^b	Key function in initiation	References
eIF3g	<i>TIF35</i> / YDR429C	EIF3G, EIF3-P42, EIF3S4, eIF3-delta, eIF3-p44	35.6	Peripheral subunit. Interacts with mRNA	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3h	N/A	EIF3H, EIF3S3, eIF3-gamma, eIF3-p40	39.9	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3i	<i>TIF34</i> / YMR146C	EIF3I, EIF3S2, PRO2242, TRIP-1, TRIP1, eIF3-beta, eIF3-p36	36.5	Peripheral subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3j	<i>HCR1</i> / YLR192C	EIF3J, EIF3S1, eIF3-alpha, eIF3-p35, p35	29.1	Peripheral subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3k	N/A	EIF3K, EIF3-p28, EIF3S12, HSPC029, M9, MSTP001, PLAC-24, PLAC24, PRO1474, PTD001, ARG134	25.1	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3l	N/A	EIF3L, EIF3EIP, EIF3S11, EIF3S6IP, HSPC021, HSPC025, MSTP005	66.7	MPN/PCI subunit. Possible binding to cap structure	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF3m	N/A	EIF3M, B5, PCID1, TANGO7, hfl-B5, GA17	42.5	MPN/PCI subunit	Cate (2017), Erzberger et al. (2014), Valasek et al. (2017)
eIF4F (composite)		eIF4, eIF-4, eIF-4F, CBP-II	251.2	Formation of mRNA closed-loop. SSU attachment to the mRNA cap, initial mRNA unwinding and loading onto SSU, possibly scanning directionality and motion	Asano (2014), Hinnebusch (2017), Jackson et al. (2010), Merrick (2015), Pelletier, Graff, Ruggero, and Sonenberg (2015)
eIF4A	<i>TIF1</i> / YKR059W <i>TIF2</i> / YJL138C	DDX2A, EIF-4A, EIF4A, eIF4A-I, eIF4A-II, eIF4A, IF-E4, IF-M4, IF _{EMC} EIF4A2, BM-010, DDX2B, EIF4A, EIF4F, eIF4A-II, eIF4A-II	46.2	ATP-dependent RNA helicase. In eIF4F, it possibly couples ATP hydrolysis with scanning directionality and motion	Garcia-Garcia, Frieda, Feoktistova, Fraser, and Block (2015), Ozes, Feoktistova, Avanzino, and Fraser (2011), Sun et al. (2014)
eIF4E	<i>CDC33</i> / YOL139C	AUTS19, CBP, EIF4E1, EIF4EL1, EIF4F, eIF-4E, CBP-I, eIF-4α	28.8	Cytoplasmic cap-binding protein. Interacts with eIF4G. Is regulated by 4E-binding proteins that block its binding to eIF4G	O'Leary, Petrov, Chen, and Puglisi (2013), Ozes et al. (2011), Papadopoulos et al. (2014)
eIF4G	<i>TIF4631</i> / YGR162W <i>TIF4632</i> / YGL049C	EIF4G1, EIF-4G1, EIF4G, EIF4G1, P220, PARK18 EIF4G3, eIF-4G 3, eIF4G 3, eIF4GII, eIF-4γ	176.2	Cap-dependent attachment of mRNA to eIF3 (mammals) or SSU (yeast). Possible coupling of ATP hydrolysis by eIF4A with scanning directionality and motion	Kumar, Hellen, and Pestova (2016), Ozes et al. (2011), Villa, Do, Hershey, and Fraser (2013), Zhao, Liu, Miller, and Goss (2017)
eIF4A	See above	See above	See above	See above	See above
eIF4B	<i>TIF3</i> / YPR163C	IF-E6, if-M3, EIF4B, EIF-4B, PRO1843	69.1	Single-stranded RNA binding protein. Possibly contributes to scanning directionality and motion	Andreou, Hams, and Klostermeier (2017), Sen, Zhou, Harris, Ingolia, and Hinnebusch (2016)
eIF4H	N/A	EIF4H, WBSCR1, WSCR1, eIF-4H	27.4	Single-stranded RNA binding protein	Sun et al. (2012)
eIF5	<i>TIF5</i> / YPR041W	eIF-5, IF-E5, IF-M2A, IF-11, IF-L2, IF-3, EIF5, EIF-5, EIF-5A	49.2	GTPase-activating protein for eIF2 GTPase. Participates in the control of P ₁ _{sc} /P _{out} switching during scanning and start codon recognition	Asano (2014), Hinnebusch (2017), Jackson et al. (2010), Obayashi et al. (2017), Tang et al. (2017)
eIF5A	<i>HYP2</i> / YEL034W <i>ANB1</i> / YJR047C	eIF-5A, eIF-4D, IF-M2Bα, EIF5A, EIF-5A, EIF5A1, eIF5A1	20.2	Stimulates peptide elongation during translation of specific codon contexts. Accelerates peptide release during termination	Rossi, Kuroshu, Zanelli, and Valentini (2014), Schuller, Wu, Dever, Buskirk, and Green (2017)

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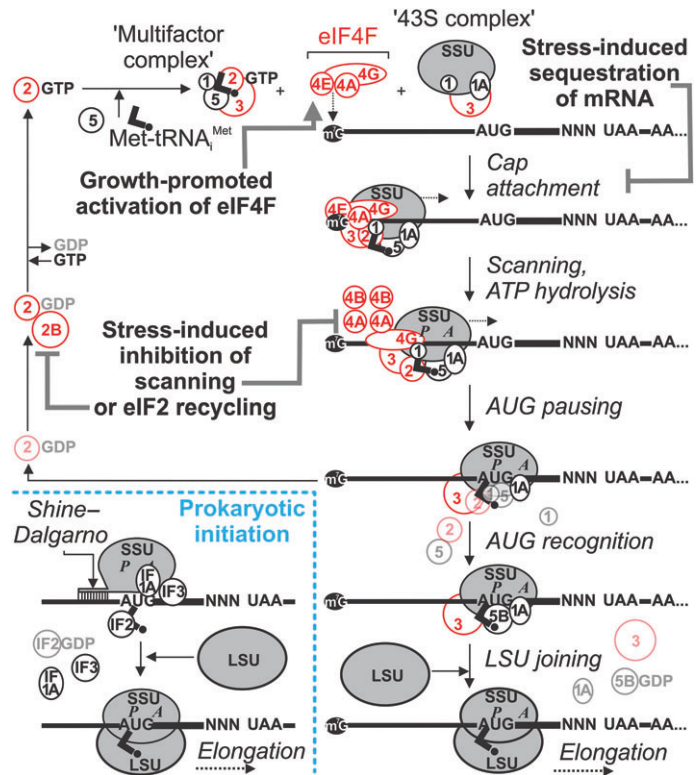
TABLE 1 (Continued)

Initiation factor	Name in yeast	Also known as	MW, kDa (human) ^b	Key function in initiation	References
eIF5B	FUN12 / YAL035W	eIF-5B, EIF5B, EIF-5B, IF2	138.8	GTP-dependent Met-tRNA _i ^{Met} positioning on the SSU and LSU during subunit joining	Asano (2014), Hinnebusch (2017), Jackson et al. (2010), Nag et al. (2016), Pisareva and Pisarev (2014)
eIF6	TIF6 / YPR016C	eIF-3A, EIF6, CAB, EIF3A, integrin beta 4 binding protein, ITGB4BP, eIF-6, p27 (BBP), p27BBP	26.6	Ribosomal subunits anti-association and biogenesis; microRNA-mediated silencing	Brina, Miluzio, Ricciardi, and Biffo (2015)
PABP	PAB1 / YER165W	Poly(A)-binding protein, PABPC1, PAB1, PABP1, PABPC2, PABPL1	70.7	Binding to poly(A) sequences at the mRNA 3' end, interaction with eIF4G, and formation of the mRNA closed-loop in capped mRNAs. Stimulates translation and possibly promotes ribosomal recycling and reinitiation	Kahvejian, Svitkin, Sukarich, MBoutchou, and Sonenberg (2005), Martineau et al. (2008)
DAP5	N/A	DAP-5, eIF4G2, EIF4G2, AAG1, NAT1, P97	102.4	Functional analogue of eIF4G for cap-independent translation initiation on certain mRNAs	Liberman et al. (2015)
MCT-1	TMA20 / YER007C-A (possible)	Multiple copies in T-cell lymphoma-1, malignant T cell-amplified sequence 1, MCT1, MCTS1	20.6	GTP-independent Met-tRNA _i ^{Met} delivery in complex with DENR on certain mRNAs and during reinitiation after uORF translation	Weisser et al. (2017)
DENR	TMA22 / YIR014W (possible)	Density regulated protein	22.3	GTP-independent Met-tRNA _i ^{Met} delivery in complex with MCT-1 on certain mRNAs and during reinitiation after uORF translation	Weisser et al. (2017)
DHX29	DED1 / YOR204W (analogue)	DDX29	155.2	ATP-dependent RNA helicase; required for scanning of 5' UTRs with highly stable hairpin structures	des Georges et al. (2015), Pisareva and Pisarev (2016)

^a Only factors that directly interact with mRNA-ribosomal complexes during cap binding, scanning, start codon recognition, and LSU joining stages are listed (see text for additional factors, most notably eIF2B).

^b Molecular weights are given only for major isoforms of the proteins.

FIGURE 2 Eukaryotic translation initiation *via* cap-dependent scanning of mRNA. Note that, as initiation is highly dynamic, there is some uncertainty regarding the exact temporal sequence of factor interactions and subcomplex assembly. The schematic presents one possible route of initiation, but other routes involving the same factors might well exist, also depending on type of mRNA and experimental system. For example, instead of parallel but separate formations of “multifactor” and “43S” complexes, the involved factors have also been proposed to assemble with each other in a different order (Aylett & Ban, 2017; Hinnebusch, 2017; Jackson et al., 2010). Late stages of eukaryotic initiation are depicted next to the homologous stages of the much simpler prokaryotic initiation process. Key cellular control inputs are further indicated in bold font. Eukaryotic translation initiation factors (eIFs), whose dysregulation prominently associates with the onset or maintaining of malignant phenotype, are shown in red (Bitterman & Polunovsky, 2012; Chu, Cargnello, Topisirovic, & Pelletier, 2016; Bramham, Jensen, & Proud, 2016; Wortham & Proud, 2015)



we further show the structures of the main eIFs and their locations on the SSU in Figure 3 (where structural data exist) and Figure 4 (where eIF locations can only be extrapolated from indirect evidence). It is useful, although somewhat arbitrary, to divide the dynamic process of cap-dependent initiation into several distinct stages. Here, we focus on mRNA-associated events and divide them, based on the activity of ribosomal complexes as well as their localization along mRNA, into three stages: (a) attachment to the cap, (b) scanning, and (c) start codon recognition including LSU joining (the latter is sometimes considered as a separate stage).

During attachment to the cap, SSU interactions with mRNA are aided by the eIF4 group of factors. The cytoplasmic cap-binding protein eIF4E (Figure 4e) interacts with the 5' cap and binds to the scaffold protein eIF4G (Figure 4c). The ATP-dependent RNA-helicase eIF4A (Figure 4a) interacts more transiently with eIF4G:eIF4E to form an eIF4F complex. eIF4F then recruits the SSU *via* eIF4G affinity to eIF1, eIF5, and possibly rRNA (yeast) or *via* eIF4G affinity to eIF3 (higher eukaryotes) (Figures 3f,g). A preformed ternary complex (TC) of the Met-tRNA_i^{Met} (Figure 3b), GTP, and eIF2 (Figure 3e; not homologous to bacterial IF2) also joins this complex, as do eIF1 (Figure 3c) and eIF1A (Figure 3d). eIF1 is analogous to prokaryotic IF3 and functions in destabilizing codon–anticodon interactions. eIF1A is a direct homologue of IF1, whose conserved functions are deemed to be that of a “placeholder” in the SSU A-site, possibly to prevent early elongator tRNA binding, and of a stabilizer of mRNA and Met-tRNA_i^{Met} binding to the SSU.

The SSU and associated factors then commence the scanning stage by moving 3'-ward on mRNA. This scanning motion can require energy expense through hydrolysis of ATP (or other nucleotide triphosphates) and is thought to be linked to the activity of RNA helicases, such as eIF4A and DHX29 (Figure 5c), assisted by the single-stranded RNA-binding protein eIF4B (and, additionally, eIF4H in higher eukaryotes) (Figure 4b).

The start codon recognition stage begins when SSUs halt their movement at a start codon appearing in the SSU P-site, surrounded by a “strong” nucleotide context. A series of rearrangements is initiated, leading to the dissociation of some of the scanning components, including eIF1 and eIF2:GDP. Met-tRNA_i^{Met}, located in the P-site, then binds eIF5B:GTP (Figure 3h), which is partially homologous to prokaryotic IF2. From here on, eukaryotic initiation strongly resembles its prokaryotic counterpart and concludes in the LSU joining and the release of eIF5B. After each initiation round, eIF2:GDP needs to be recycled by its GDP/GTP exchange factor eIF2B (Bogorad, Lin, & Marintchev, 2017; Jennings & Pavitt, 2014; Wortham & Proud, 2015). This eIF2 recycling is highly regulated and generally considered an additional stage of initiation that takes place away from the ribosome and mRNA (Wortham & Proud, 2015).

The cap-and-scanning initiation pathway, as summarized above, can be seen as universally conserved across eukaryotes, as are the main initiation factor sets (Browning & Bailey-Serres, 2015). Nevertheless, the prevalence of isoforms for certain eIFs and their regions with strong homology can vary substantially between taxa, especially for the group 4 eIFs. For example, in higher plants, additional eIF(iso)4E and eIF(iso)4G can form alternative and well-represented eIF(iso)4F complexes

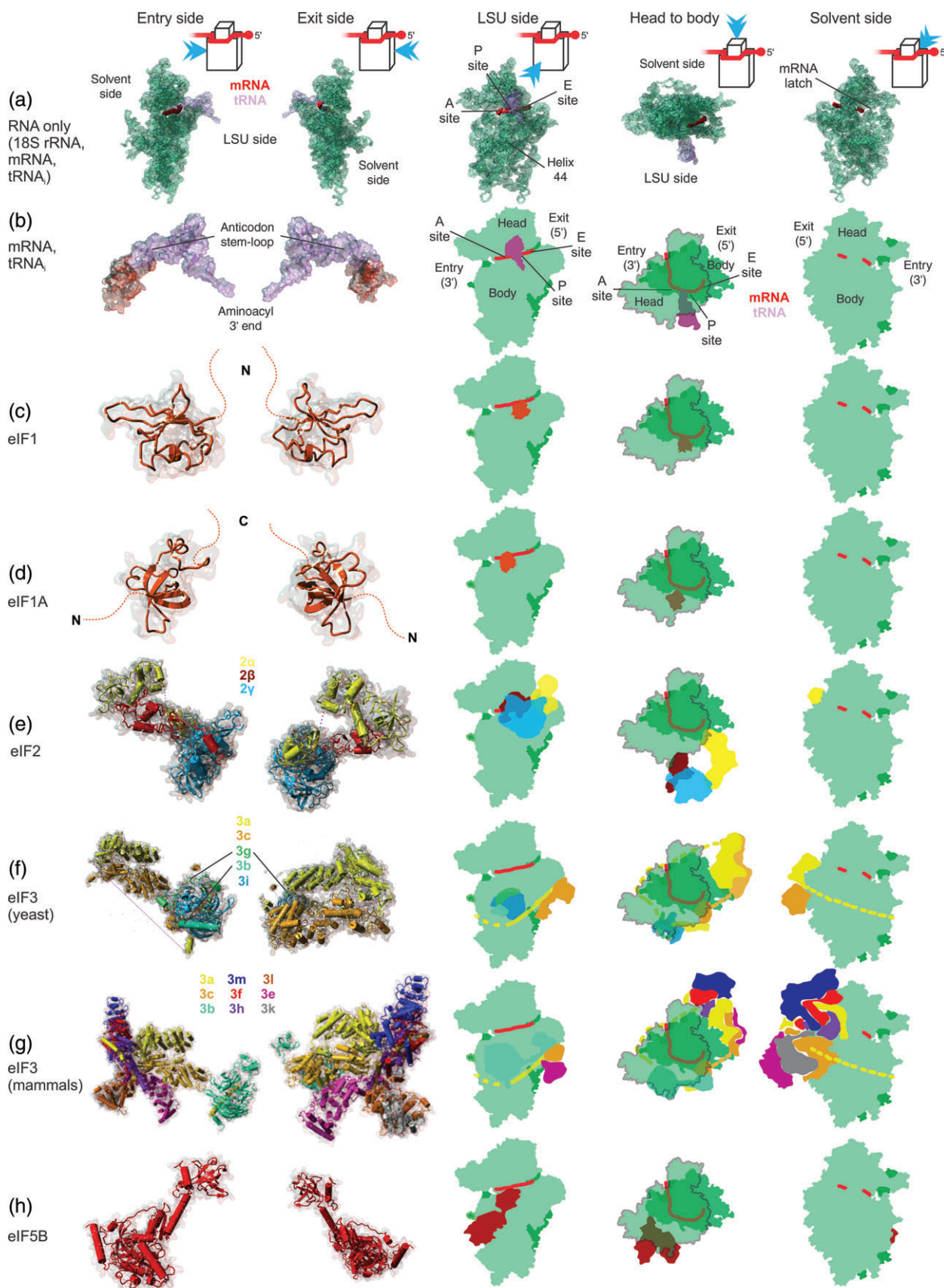


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that target specific mRNA subsets and locations (Browning & Bailey-Serres, 2015; Kropiwnicka et al., 2015; Martinez-Silva, Aguirre-Martinez, Flores-Tinoco, Alejandri-Ramirez, & Dinkova, 2012; Monzingo et al., 2007), whereas among the variety of trypanosome eIF4E homologues (~6) (Freire, Sturm, Campbell, & de Melo Neto, 2017), eIF4E4 appears as the major active isoform (Klein, Terrao, Inchaustegui Gil, & Clayton, 2015). Much is still unknown about how this eIF4E variety contributes to possible selectivity of mRNA translation (Clayton, 2016; Smircich et al., 2015). Importantly, recent structural findings suggest that each of the main initiation stages corresponds to pivotal changes in the SSU conformation and its reconfiguring by the binding and release of initiation factors.

Translational regulation during the initiation phase is widely utilized in processes important for multicellular organisms, such as differentiation, cell proliferation, and site-specific translation (Besse & Ephrussi, 2008; Jansen, Niessing, Baumann, & Feldbrugge, 2014; Weill et al., 2012); it further features commonly in pathologies. For example, dysregulation of most of the eukaryote-specific eIFs is implicated in the onset of malignancies and is regarded as one of the drivers of oncogenesis (Bhat et al., 2015; Hershey, 2010). Many eIFs are targets for regulation, especially in higher eukaryotes (Hershey, Sonenberg, & Mathews, 2012; Jackson et al., 2010). Examples of such control include phosphorylation of eIF4E-binding proteins (4E-BPs) by mTOR (resulting in the release of “free” eIF4E and increased eIF4F levels); eIF4E by MNK1/2 (which may be stimulatory to translation); and eIF2 α by GCN2, HRK, PKR, and PERK (which results in the stable binding of eIF2 to eIF2B, decreased levels of TC, and skipping of upstream (u)ORFs during reinitiation of scanning) (Proud, 2011, 2015; Sonenberg & Hinnebusch, 2009; Topisirovic, Svitkin, Sonenberg, & Shatkin, 2011). Most of the other eIFs can be phosphorylated as well (Andaya, Villa, Jia, Fraser, & Leary, 2014; Jackson et al., 2010). Finally, ribosomes can also be phosphorylated (*e.g.*, SSUs at RSP6, which may have functional implications in malignant phenotypes) (Meyuhas, 2015; Wittenberg et al., 2016) or have an outright different composition, possibly affecting their specific interactions with mRNA (Preiss, 2016; Slavov, Semrau, Airoidi, Budnik, & van Oudenaarden, 2015). For example, translation of the homeobox protein-coding (Hox) mRNAs was shown to be specifically dependent on the LSU protein RPL38 (Xue et al., 2015).

It should be noted that in addition to their diversity across taxa as discussed above, many of the mammalian eIFs (especially the eIF4 group) occur as different isoforms and/or paralogues that may function in special modes of initiation and its control. The eIF4E paralogues eIF4E1, -4E2, -4E3 and their isoforms are known to differentially control mRNA localization to stress granules and processing bodies (Frydryskova et al., 2016). eIF4G3 (eIF4GII) (divergent from eIF4G1 [eIF4GI] in the eIF4E-binding N-terminal part) was suggested to have a distinct effect on translation (Coldwell et al., 2012; Ghosh & Lasko, 2015). eIF4GII (but not eIF4GI) cleavage by viral proteases blocks the translation of host proteins (Svitkin, Gradi, Imataka, Morino, & Sonenberg, 1999). DAP5 (or eIF4G2, p97, NAT1) (Figure 4c) is ~70% homologous to the middle and C-terminal parts of eIF4GI but lacks the N-terminal eIF4E and PABP binding sites (Figure 4f); it can thus stimulate cap-independent translation (Nousch, Reed, Bryson-Richardson, Currie, & Preiss, 2007; Lewis et al., 2008; Liberman et al., 2015; Sugiyama et al., 2017). Although eIFs 4A1 and 4A2 are highly homologous, they are functionally noninterchangeable (Lu, Wilczynska, Smith, & Bushell, 2014). The more divergent eIF4A3 protein is not an initiation factor (Lu et al., 2014). Prominent examples of eIFs important during developmental processes include DAP5, whose knockout results in the differentiation-resistant mouse embryonic stem cells (while their transcriptional profiles remain the same) (Sugiyama et al., 2017); eIF4B, which can control survival and proliferation of the cells (Shahbazian, Parsyan, Petroulakis, Hershey, & Sonenberg, 2010); eIF4E, whose liberation from eIF4E–BP interactions (Kamenska, Simpson, & Standart, 2014) coincides with oocyte maturation in mice (Han et al., 2006) and whose intracellular location-dependent inhibition defines embryonic

FIGURE 3 Structures and ribosomal localization (directly available from structural data) of the main components of cap-dependent translation initiation. Cartoons on the top indicate the point of view relative to the SSU oriented head up, intersubunit side front, and mRNA entry side (opposite to the 5' cap) to the left. Surfaces are calculated and outlines are derived from published structural data. Rendering was performed with the YASARA (Krieger & Vriend, 2014) Model. (a) Arrangement of the SSU rRNA (green), P-site Met-tRNA_i^{Met} (purple), and mRNA (red) (Lomakin & Steitz, 2013). (b) Views of Met-tRNA_i^{Met}–mRNA interactions at the P-site of the initiating SSU (left) (Lomakin & Steitz, 2013) and schematics of the mRNA and Met-tRNA_i^{Met} location on the initiating SSU with key structural features of the SSUs indicated in outline (right) (Llacer et al., 2015). In the head-to-body SSU view, the head is rendered semitransparent to allow observation of the body outline, and the head outline is augmented by a gray bezel line. The darker green areas in all three views (LSU side, head-to-body, solvent side) indicate SSU portions that are located to the rear of the main outline plane to improve depth perception. (c) Secondary structure and surface of the globular ribosome-bound eIF1 core (Lomakin & Steitz, 2013) with its N-terminal unstructured extension indicated by the dotted line (left) and outline of its location on the SSU (right) (Llacer et al., 2015). (d) Same as (c) but for eIF1A (Lomakin & Steitz, 2013). (e) Secondary structure and surface of yeast eIF2 as part of the SSU-bound TC (left) and its outline on the SSU (right) (Llacer et al., 2015). (f) Same as (e) but for yeast eIF3 (Llacer et al., 2015). (g) Structure of mammalian eIF3 from the eIF3:SSU mRNA-free complex also containing DHX29 (des Georges et al., 2015) shown with yeast SSU outline and aligned by its eIF3c subunit to the yeast eIF3c in the eIF3:SSU complex) (Llacer et al., 2015) using MUSTANG (Konagurthu, Whisstock, Stuckey, & Lesk, 2006). eIF3a and eIF3c intersubunit exit-side extensions repeated from yeast structure in (f). eIF3b manually docked from the mammalian eIF3 as part of the mRNA-bound SSU complex (Simonetti et al., 2016). (h) Secondary structure and surface of rabbit eIF5B as part of the eIF5B:GMP-PNP:Met-tRNA_i^{Met} complex bound to the complete ribosomes in complex with the hepatitis C virus (HCV) internal ribosome entry site (IRES) (left) (Yamamoto et al., 2014) and its outline on the yeast SSU (right) (Llacer et al., 2015). Alignment of the yeast and rabbit structures was performed by the 18S rRNA trace in the YASARA Model

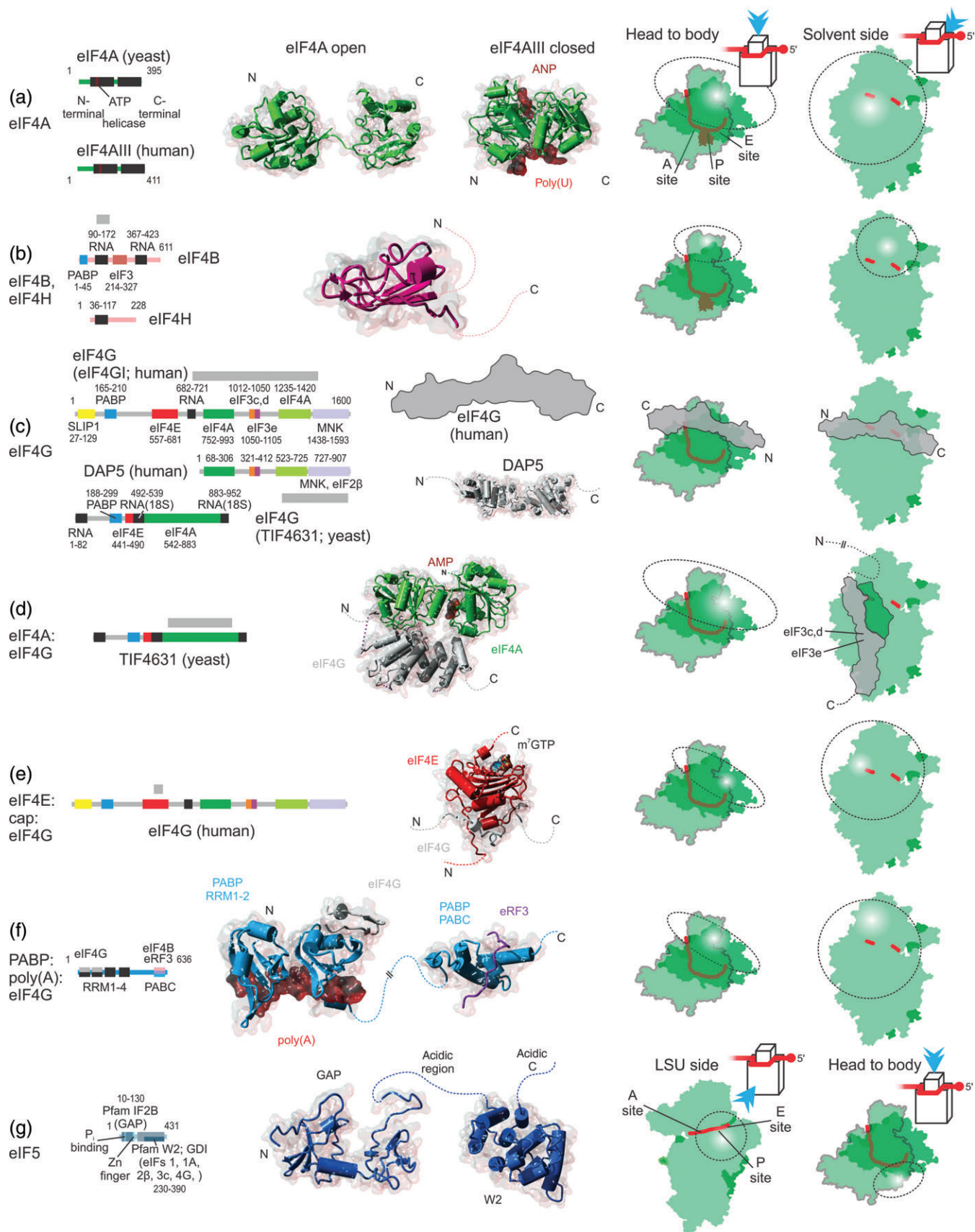


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patterning in *Drosophila* (Castagnetti, Hentze, Ephrussi, & Gebauer, 2000; Yarunin, Harris, Ashe, & Ashe, 2011), synaptic plasticity and memory (Weill et al., 2012), pre-programming of oocytes (Jansen et al., 2014; Minshall, Reiter, Weil, & Standart, 2007), and regulation of daughter cells growth in yeast (Besse & Ephrussi, 2008); RNA helicases of translation such as DDX3Y, which is essential for spermatogenesis; eIF3 subunit a, which can control translation of important cell cycle regulators and cell entry into the S-phase (Saletta, Suryo Rahmanto, & Richardson, 2010); eIF3 subunit f, which is critical for maintenance of skeletal muscle size during starvation (Sanchez et al., 2013), eIF3 subunit h, which directs translation of specific transcripts required for development of brain and eyes in zebrafish (Choudhuri, Maitra, & Evans, 2013); and others (Abdelhaleem, 2005; Kang & Han, 2011; Richter & Lasko, 2011; Standart & Minshall, 2008). Many altered eIFs and their regulators are direct causes of inherited disorders, such as childhood ataxia (eIF2B), diet-induced obesity (eIF2 α), diabetes (eIF2 α kinase EIF2AK3/PERK), and others (Scheper, van der Knaap, & Proud, 2007).

In addition, eukaryotic mRNAs richly interact with RNA-binding proteins (RBPs), dynamically forming messenger ribonucleoproteins (mRNPs) to control their life cycle, from their production in the nucleus, export into the cytoplasm, localization, and translation to their eventual decay (Mitchell & Parker, 2014). RBPs can be abundant and universal, such as the cap-binding proteins, the major mRNP constituent Y-box binding protein 1 (YB-1) (Skabkin et al., 2004), and poly(A)-binding proteins, or mRNA-specific (Castello et al., 2012; Hentze, Castello, Schwarzl, & Preiss, 2018). The diversity and dynamic nature of mRNPs and the added complexity of translation initiation in eukaryotes allow for a richness of post-transcriptional gene control. Indeed, protein synthesis rates have been shown to differ by up to ~1,000-fold between eukaryotic mRNAs (Schwanhauser et al., 2011). The literature abounds with mechanisms to alter translation initiation in an mRNA-selective way, *e.g.*, through *cis*-acting regulatory motifs in the mRNA UTRs (Hershey et al., 2012; Hinnebusch et al., 2016; Jackson et al., 2010). Scanning “roadblocks” in 5′ UTRs are well recognized as the basis of translational control (Hinnebusch et al., 2016). This is exemplified by uORF regulatory effects of *GCN4* mRNA in yeast or ATF4 mRNA in mammals, which inhibit main ORF translation when the levels of active eIF2 are high (Harding et al., 2000; Hinnebusch et al., 2004), and by the binding of iron-regulatory protein to the ferritin mRNA 5′ UTR, which leads to the steric hindrance of scanning when the cytoplasmic iron concentration is low (Gebauer & Hentze, 2004). Recent insights gained with ribosome profiling approaches confirm widespread effects of 5′ UTR uORFs or upstream start codons in the control of canonical initiation (Andreev et al., 2017). The 3′ UTRs are also often involved in the control of cap-dependent attachment *in cis*, specifically by binding proteins that form a stable closed-loop of mRNA but cannot attach eIF3 or eIF4G (Gebauer & Hentze, 2004; Gebauer, Preiss, & Hentze, 2012; Jackson et al., 2010). The 3′ UTR-binding microRNAs may repress translation at different stages of initiation (Fabian, Sonenberg, & Filipowicz, 2010; Humphreys, Westman, Martin, & Preiss, 2005; Jackson et al., 2010). However, activators of scanning have also been found in 3′ UTRs, with 3′-proximally located “cap-independent translational enhancers” (CITEs) shown to specifically bind components of the cap-dependent initiation and then assist translation similar to the action of the canonical closed-loop of mRNA (Shatsky, Dmitriev, Terenin, & Andreev, 2010).

While translation of cellular mRNAs is thought to be mostly initiated through cap-dependent scanning, there are multiple documented pathways of initiation that do not fully fit into this model. Internal ribosome attachment to mRNA has been demonstrated for many cases of initiation on viral mRNAs (Johnson, Grosely, Petrov, & Puglisi, 2017; Pisarev, Shirokikh, &

FIGURE 4 Structures and schematics of possible ribosomal localization of the main components of cap-dependent translation initiation for which no precise SSU position is known. SSU depiction as in Figure 3. The likely SSU localization is shown as a white semitransparent circle (unless stated otherwise), with a wider region of high probability of localization indicated by a dashed black line (right; on the SSU outline). (a) Domain structures of yeast eIF4A and human eIF4AIII (eIF4AI homologue) (left). Crystal structure of yeast eIF4A in the open conformation (Caruthers, Johnson, & McKay, 2000) and human eIF4AIII in the closed conformation bound to single-stranded poly(U) RNA and AMP-PNP (ANP) (Buchwald et al., 2010) (middle). (b) Domain structures of human eIF4B and its partial homologue eIF4H (aligned by the region of homology; left). Solution structure of the human eIF4B amino acids 96–176 (shown by the gray bar in the human eIF4B domain scheme on the left) (middle) (Fleming et al., 2003). Position of eIF4B on the SSU outline accounting for directed hydroxyl radical mapping data (Walker et al., 2013). (c) Domain structures of human eIF4G and DAP5 (eIF4G2, p97, NAT1, AAG1) and yeast eIF4G (TIF4631) (left). Outline of the small angle X-ray scattering (SAXS) envelope of human eIF4G amino acids 712–1,451 (shown by the gray bar in the human eIF4G domain scheme on the left) (Nielsen et al., 2011) with the eIF4A-corresponding density cut out manually (middle top). Crystal structure of the N-terminal part of human DAP5 (Fan, Jia, & Gong, 2010) (shown by the gray bar in the human DAP5 domain scheme on the left) (middle bottom). eIF4G SAXS envelope (minus eIF4A) shown in-scale with the SSU and positioned as originally proposed (Nielsen et al., 2011). (d) Crystal structure of yeast eIF4G eIF4A-binding domain amino acids 571–853 (shown by the gray bar in the eIF4G domain scheme on the left) bound to yeast eIF4A:AMP complex (middle) (Schutz et al., 2008). Outline of the human eIF4A:eIF4G SAXS envelope (see (c)) (Nielsen et al., 2011) shown in scale with the SSU and oriented by eIF4G eIF3 binding sites to the locations of eIF3c, d, e subunits and the position of the functionally analogous HCV IRES (see text) (right) (Yamamoto, Unbehaun, & Spahn, 2017). (e) Domain scheme (left) and crystal structure of human eIF4E bound to human eIF4G amino acids 608–642 (middle) (Gruner et al., 2016). (f) Domain scheme (left) and combined crystal structures of the human PABP RNA recognition motif (RRM) 1–2 domains bound to (A)₉ RNA and amino acids 179–198 of human eIF4G (Safaei et al., 2012), together with PABP C-terminal (PABC) domain bound to amino acids 59–73 of human eIF4B (middle) (Kozlov, Menade, Rosenauer, Nguyen, & Gehring, 2010). (g) Domain scheme (left) and combined solution structure of the N-terminal GTPase-activating protein (GAP) domain (Conte et al., 2006), together with crystal structure of the C-terminal Pfam W2 domain (middle) (Bieniossek et al., 2006), of human eIF5. Possible localization on the ribosome taking into account cryo-EM data (right) (Hussain et al., 2014)

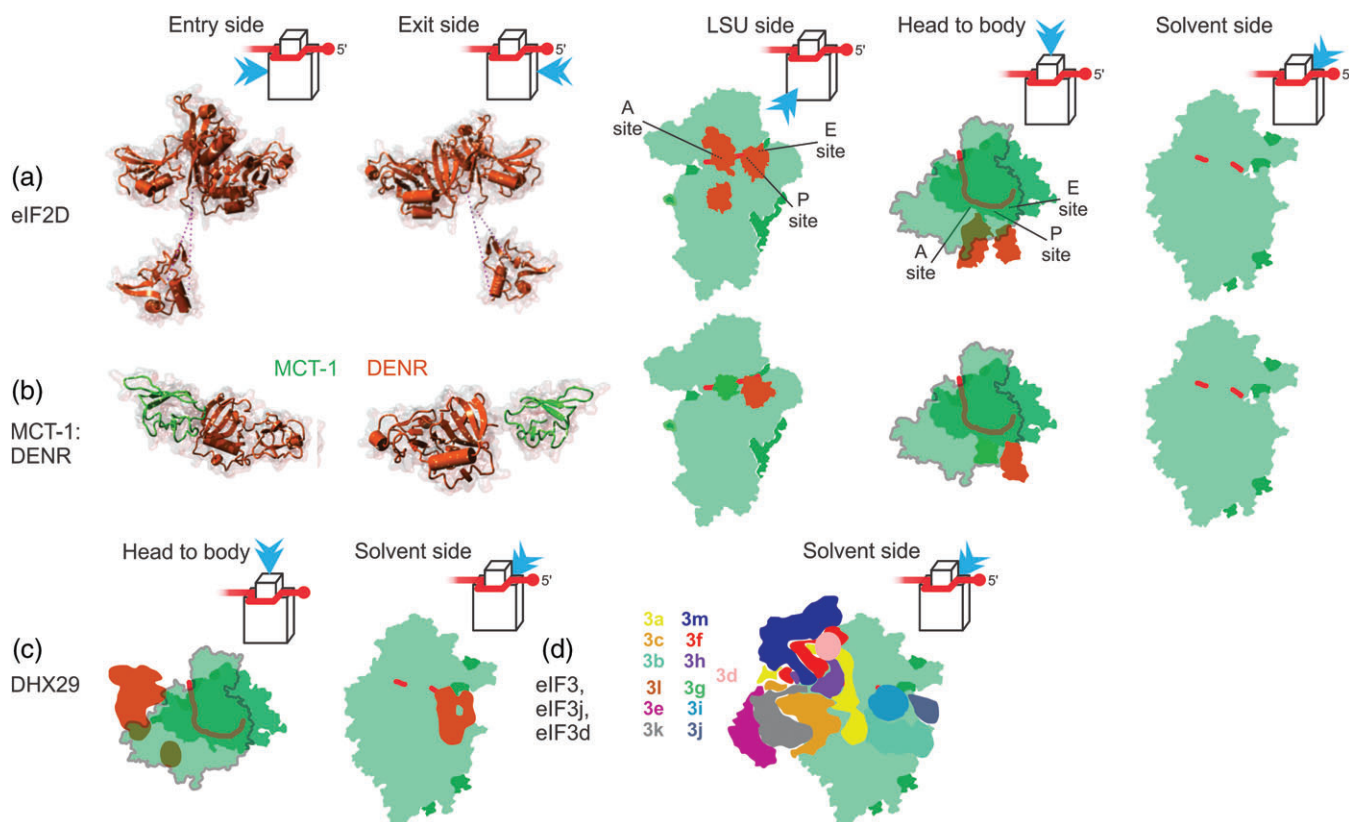


FIGURE 5 Structures and schematics of ribosomal localization of some of the additional initiation components, including an additional configuration of eIF3, with eIF3j in the entry of the mRNA-binding cleft of the mRNA-free SSUs. Designations and SSU schematics are as in Figure 3. (a) Cryo-EM structure of human eIF2D bound to the human SSU (left) (Weisser et al., 2017). Outline of the human eIF2D on yeast SSU (SSU outline as described in Figure 3); alignment of structures performed by the 18S rRNA trace (right). (b) Crystal structures of human density regulated reinitiation and release factor (DENR; C-terminal half, amino acids 110–198) and multiple copies in T-cell lymphoma-1 (MCT-1) bound to the human SSU (left) (Lomakin et al., 2017) and the outline of these factors shown on yeast SSU aligned to the human SSU as in (a) (right). (c) Cartoon of the cryo-EM density localization of rabbit DHX29 in complex with SSU, eIFs 1, 1A, GMP-PNP:2:Met-tRNA^{Met}, and 3 modified from the original publication (Hashem et al., 2013a) and shown on the yeast SSU outline. (d) Cartoon of eIF3j, eIF3i, and eIF3b positions on the solvent SSU side near mRNA entry as observed in the mRNA-free “43S” complexes (Erzberger et al., 2014). A tentative localization of eIF3d shown as a pink circle. Modified from the original publication (Erzberger et al., 2014) and shown on the yeast SSU outline and the other eIF3 subunits as in Figure 3g

Hellen, 2005; Terenin, Smirnova, Andreev, Dmitriev, & Shatsky, 2017; Yamamoto et al., 2017), as well as for synthetic (C. Y. Chen & Sarnow, 1995) and naturally occurring circular RNAs (Legnini et al., 2017; Pamudurti et al., 2017). In viral internal ribosome entry sites (IRESes) of types 1–3, specially folded RNA structures functionally (and to some extent structurally) mimic eIFs and abrogate at least the dependency on eIF4E (or complete eIF4F) and cap (IRES type 1, full scanning; type 2, limited scanning) or eIF4G and all other group 4 factors (type 3, no scanning). IRESes were also identified for a multitude of cellular mRNAs, although the mechanism(s) of their action remain(s) largely obscure due to the lack of structural information (Faye & Holcik, 2015; Jackson, 2013; Walters & Thompson, 2016). For prominent examples such as the X-linked inhibitor of apoptosis (XIAP) mRNA IRES, independence on eIF2 α and direct binding of eIF3 were reported (Thakor et al., 2017; Thakor & Holcik, 2012). Similar to the 3' CITEs, 5' CITEs may function by enhancing the binding of canonical initiation factors, rendering translation less dependent on the 5' cap while preserving overall 5'-end dependence of initiation. The 5' CITEs were suggested as a more conservative explanation of the cellular IRES phenomenon on some mRNAs, such as apoptotic protease-activating factor (Apaf)-1 mRNA (Shatsky et al., 2010). The above-mentioned Hox mRNA translation functionally demonstrates internal attachment of ribosomes, with cap-dependent initiation seemingly specifically suppressed (Xue et al., 2015). Recent high-throughput studies facilitated characterization of initiation by various other “noncanonical” routes. mTOR-dependent regulation of mRNAs with 5'-proximal “terminal oligopyrimidine” (TOP) sequences (characteristic of many mRNAs encoding eIFs) was identified using ribosome profiling (Thoreen et al., 2012); further investigations detailed the existence of non-TOP mRNAs with similar properties (Gandin et al., 2016). G-quadruplexes in 5' UTRs are often linked to IRES activity (Bhattacharyya, Diamond, & Basu, 2015) and, at the same time, were demonstrated as prominent determinants of eIF4A dependency genome-wide (Wolfe et al., 2014). G-quadruplexes are present in 5' UTRs of many oncogenic mRNAs and link elevated eIF4A activity with cancer phenotypes (Wolfe et al., 2014). eIF2A-dependent translation of the phosphatase and tensin homologue (PTEN, a notable tumor suppressor) mRNA is initiated noncanonically from

the CUG codon and leads to the production of the PTEN α isoform (Liang et al., 2014). Synthesis of PTEN α from PTEN mRNA requires the CUG start codon to be placed in a 16 nt palindromic motif CCCGCUCCUGGAGCGGG, but the details of the mechanism remain unknown (Liang et al., 2014). Interestingly, repeat-associated non-AUG (RAN) translation, typical to many repeat expansion disorders, was recently demonstrated to utilize CUG start codons (Green et al., 2017; Green, Linsalata, & Todd, 2016).

A paradigm-shifting discovery was the widespread mapping of N^6 -methyladenosine-modified nucleotides (m^6A) in eukaryotic mRNA and the characterization of their action in the 5' UTR as internal ribosome attachment points (Meyer et al., 2015). m^6A was demonstrated to directly bind eIF3 (skipping cap dependence and eIF4E action) and promote the translation of oncogenic mRNAs (Lin, Choe, Du, Triboulet, & Gregory, 2016; Meyer et al., 2015). Sometimes, the cap-dependent scanning pathway is antagonized by internal or cap-independent initiation (Kozak, 2001, 2003; Lacerda, Menezes, & Romao, 2017; Schneider et al., 2001). However, these “alternative” pathways might not be entirely in conflict with the cap-dependent scanning model. In most “noncanonical” cases, including many of the internal attachment events, elements of (possibly limited) scanning are still in play. Similarly, even though reduced, a (sub)set of “canonical” scanning factors is still utilized in accordance with their established functions (Shatsky et al., 2010). Thus, these other routes of initiation can be viewed as more or less profoundly modified versions of the “canonical” initiation pathway (Sweeney, Abaeva, Pestova, & Hellen, 2014). There are some truly different established mechanisms, such as initiation at the intergenic regions of dicistroviruses (IRES type 4, an elongation-derived mechanism) (Pisarev et al., 2005), eIF2-independent delivery of tRNA (Dmitriev et al., 2010; Pestova, de Breyne, Pisarev, Abaeva, & Hellen, 2008; Skabkin et al., 2010), initiation with leucine elongator tRNA by the alternative tRNA carrier eIF2A at CUG start codons (Starck et al., 2016), or initiation on “leaderless” mRNA by complete ribosomes (Akulich et al., 2016). Some principal alternatives to Kozak's scanning (such as initiation by mRNA “looping”) have also been proposed (Jang & Paek, 2016); however, these are outside the focus of this review.

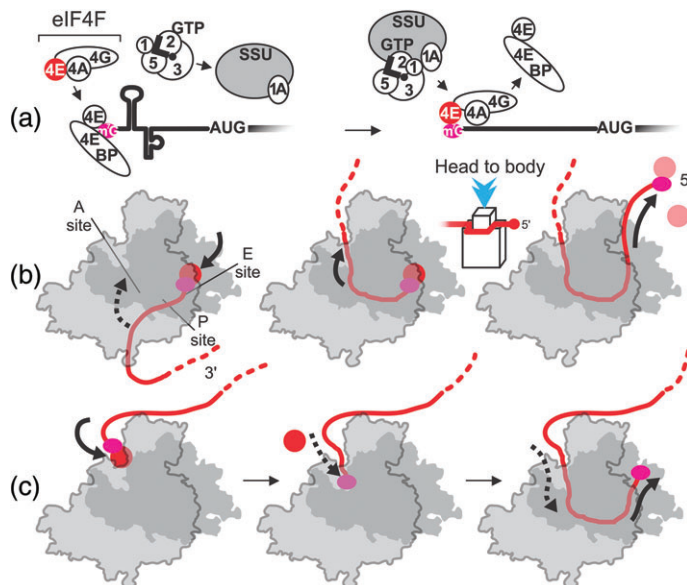
3 | ATTACHMENT TO THE CAP

Canonical translation initiation begins with cap-dependent recruitment of the SSU to the mRNA near the 5' end (Figure 1a). As the SSU has no intrinsic affinity for the cap, this requires a subset of eIFs to bridge between the SSU and the cap (and the 3' poly(A) as per the closed-loop model). eIFs required for subsequent scanning are also recruited at this early stage. A possible model for these steps is depicted in the upper part of Figure 2 and is described in detail below; however, there are also probable alternative scenarios and open questions due to the existence of alternative modes of cap recognition and a difficulty of experimentally determining the exact sequence of events.

3.1 | Assembly of translation components at the mRNA cap

An initial step is the attachment of eIF4F to the mRNA cap, which is hypothesized to unfold structures near the 5' end of mRNA, making it capable of entering the mRNA-binding cleft of the initiating SSU in single-stranded form. The 3' poly(A) assists in eIF4F recruitment through contact between PABP and eIF4G. SSU recruitment is then brought about by a further

FIGURE 6 Alternative models for cap-dependent SSU attachment to mRNA. (a) Schematic of interactions between SSU and eIFs at the cap structure. (b) Model of lateral mRNA attachment to the SSU. eIF4E (red) bound to mRNA cap (purple) first attaches near the mRNA exit site, at the solvent side of the SSU near the head-to-body junction. The 3'-ward portion of mRNA is then “slotted” laterally into the mRNA binding cleft. Once mRNA is stably bound, eIF4E:SSU and likely eIF4E:mRNA interactions are broken as a result of rearrangements upon commencement of scanning, as depicted by possible cap-bound or “free” eIF4E molecules (red, semitransparent). Solid arrows indicate completed, and dashed – continuing or anticipated, movements or processes. (c) Model of mRNA “extrusion” into the SSU (color scheme and indications as in (b)). Cap-bound eIF4E first attaches near the SSU mRNA entry. The mRNA then detaches from eIF4E and is sequentially relocated (pushed) into the SSU mRNA-binding cleft, first reaching the A-site and continuing to P- and E-sites, and eventually to the mRNA exit site of the SSU. In this case, eIF4E is required to dissociate from the cap structure after mRNA attachment to the SSU. SSU schematics as in Figure 3



interaction of eIF4G with SSU-bound eIF3 (in mammals [Weber, Hickey, & Baglioni, 1978]), which may also be mediated by eIF5 and eIF1 (as there are no eIF3-interacting regions in eIF4G in yeast (Asano et al., 2001; He et al., 2003; Hinnebusch, 2014). There is a possibility of direct eIF4G:SSU interaction *via* binding to rRNA in yeast (Berset, Zurbriggen, Djafarzadeh, Altmann, & Trachsel, 2003). Yeast eIF4E is an abundant cytoplasmic protein and could thus occupy mRNA caps for most of the time *in vivo* (von der Haar, Gross, Wagner, & McCarthy, 2004). In mammals, the eIF4E-to-eIF4G ratio can vary depending on conditions and cell type, due to altered abundance of the factors (Avdulov et al., 2015) or regulation of their activity. eIF4E is also known to have a higher affinity for the capped mRNA as part of the eIF4F complex (Gross et al., 2003; Ptushkina et al., 1998; Yanagiya et al., 2009), which can be further positively modulated by eIF4G–PABP interactions (Borman, Michel, & Kean, 2000; O'Leary et al., 2013; Wei, Balasta, Ren, & Goss, 1998). However, eIF4E interactions with eIF4E-binding proteins (4E-BPs), competitive inhibitors of eIF4G binding, have also been shown to increase cap binding (Ptushkina, von der Haar, Karim, Hughes, & McCarthy, 1999). Thus, “free” eIF4F could displace eIF4E (or eIF4E:4E-BP) from the cap as the first step of initiation, potentially with the assistance of mRNA closed-loop, for many of the steady-state initiation events (Figure 6a). Alternatively, eIF4F could bind the cap as part of an SSU complex as eIF4F can be efficiently isolated from high-salt ribosomal washes but not from ribosome-free cytosolic fractions (Duncan, Milburn, & Hershey, 1987; Hershey, Yanov, & Fakunding, 1979).

Other factors can bind the cap in the cytoplasm to stimulate or inhibit translation. At least during the initial rounds of initiation, the predominantly nuclear cap-binding complex (CBC) (Maquat, Tarn, & Isken, 2010) can be in play; it can also interact with eIF4G and support initiation, although the CBC:eIF4G interaction was found to be dispensable in yeast (Baron-Benhamou, Fortes, Inada, Preiss, & Hentze, 2003) unlike in mammals (Lejeune, Ranganathan, & Maquat, 2004). Upon thermosensitive eIF4E inactivation in yeast, CBC fails to support high levels of translation (Altmann, Sonenberg, & Trachsel, 1989); the residual low levels of translation of select mRNAs are nevertheless sufficient to support cells in a non-dividing state (Paz & Choder, 2001). Metazoans have additional eIF4E-like variants expressed from eIF4E2 (also known as 4EHP) and eIF4E3 genes, which are distinguished by their inhibitory action on translation. 4EHP binds cap, although its cap affinity is relatively low (Zuberek et al., 2007). While 4EHP and eIF4E1 can both bind the same nuclear transporter, eIF4E-T, 4EHP is not dependent on it for transport and does not revert to cytoplasmic processing bodies or stress granules during cell stress (Kubacka et al., 2013). 4EHP can bind growth factor receptor-bound protein 10-interacting glycine–tyrosine–phenylalanine domain proteins 1 and 2 (GIGYF1/2) to form stable translation suppression complexes, which prevent excessive translation (and lethality) in mouse embryos (Morita et al., 2012; Peter et al., 2017). Its cap-binding activity may be increased by binding to eIF4E-T, and it was also shown to increase microRNA-directed translation inhibition by the carbon catabolite repression 4-negative on TATA-less (CCR4-NOT) complex (Chapat et al., 2017). While a conditional knockout of eIF4E3 in mice showed no pronounced effects (van der Weyden, White, Adams, & Logan, 2011), eIF4E3 was demonstrated to bind the cap in a specialized manner, which excludes hydrophobic interactions but augments other interactions (electrostatic and van der Waals) (Osborne et al., 2013). eIF4E3 may be a direct eIF4E (eIF4E1) competitor, alleviating the stimulatory effects of eIF4E on certain mRNAs, thus preventing excessive prosurvival and proliferation-inducing activity of eIF4E, effectively being an oncosuppressor (Osborne et al., 2013). Finally, as recently discovered, eIF3 can also bind directly to the cap, through interactions with its eIF3d subunit (Lee, Kranzusch, Doudna, & Cate, 2016) and/or also possibly *via* the eIF3l subunit (Kumar et al., 2016). In the case of eIF3d, the presence of a specific stem loop structure in mRNA induces the “open” state of eIF3d, rendering it competent for cap binding (Lee et al., 2016). Conversely, removal of eIF3 from this stem loop during scanning will likely break the eIF3:cap interaction due to a loss of eIF3d affinity to the cap (see also below).

Assembly of a “multi-factor complex” (MFC; eIF1:[GTP:eIF2:Met-tRNA^{Met}]:eIF3:eIF5) away from the ribosome and mRNA has been described as another early initiation substep in yeast (Jivotovskaya, Valasek, Hinnebusch, & Nielsen, 2006; L. Valasek, Nielsen, & Hinnebusch, 2002) and in mammals (Sokabe, Fraser, & Hershey, 2012). The MFC then binds to the SSU to form a “43S” complex or “native” SSU, which then attaches to eIF4F at the cap (Figure 6a) (Sokabe et al., 2012). However, it is not evident that such a sequence of events represents the predominant initiation route. Strong direct SSU interactions of eIF3j at the mRNA entry and of eIF1 in the P-site, and of eIF1 with the N-terminal part of eIF3c near the mRNA exit, were proposed to explain the binding of MFC to the SSU (Erzberger et al., 2014). The suggested role of MFC in producing a more “open” configuration of the SSU capable for mRNA attachment supports this sequence of events (Gilbert et al., 2007). However, some abundant components such as eIF1 and eIF1A can also bind the SSU independently of the MFC (Passmore et al., 2007). Furthermore, the assembly and loading of the MFC (and TC) is not mandatory to commence the initiation process *per se*. This follows from observations of reinitiation following translation of uORFs. For example, following uORF1 translation within the 5' UTR of *GCN4* mRNA in yeast, post-termination SSUs remain attached to the mRNA and resume scanning in the absence of TC (and consequently, MFC) (Gunisova, Beznoskova, Mohammad, Vlckova, & Valasek, 2016; Gunisova & Valasek, 2014; Hinnebusch, 2005). From *in vitro* experiments with purified factors, we know that the presence of eIF5 (Figure 4g) is not required for cap attachment and scanning, and it can be added as late as after start

codon recognition (Pestova et al., 2000; Unbehauen, Borukhov, Hellen, & Pestova, 2004). Met-tRNA_i^{Met} can be bound to the MFC when it loads onto the SSU, or it can join later, individually or as a part of the TC (Sokabe et al., 2012). The particular timing and extent of attachment of “general purpose” initiation factors that bind the SSU less tightly or are not ubiquitous, such as eIF4B, eIF4H, DDX3, and DHX29, are also not known. It may well be that, in live cells, there is no single sequence of factor-binding events during cap-dependent SSU attachment that is applicable to all mRNAs and in all situations (similar to early proposals for prokaryotic initiation) (Gualerzi, Risuleo, & Pon, 1977); this would have implications for mRNA-specific control of translation. Transcriptome-wide *in vivo* evidence is needed to address these uncertainties.

3.2 | Loading of SSUs on mRNA

The apparent processivity of scanning and the precision of start codon recognition imply that mRNAs cannot dissociate from scanning complexes with ease, suggesting a locking mechanism that is permissible to the linear sliding of mRNA. One of the first suggested explanations was a “threading” mechanism of mRNA attachment, where the mRNA 5′ end is inserted through a ring-shaped opening in the initiating SSU (Kozak, 1979, 1992). However, the discovery of IRESes and m⁶A-mediated initiation, the cap-binding activity of eIF3d, and the translation of naturally occurring circRNAs all indicate a more lateral loading of mRNA. It can be argued that eIF4E is located near the SSU platform, close to its E-site and toward the solvent side of the SSU (Figure 4e; further discussed in the section on scanning below). This view is based on the location of the 5′ (outbound) portion of less-flexible mRNAs in SSU complexes such as those involving IRESes (Hashem et al., 2013b; Siridechadilok, Fraser, Hall, Doudna, & Nogales, 2005; Yamamoto et al., 2014); the crosslinking of upstream mRNA portions with eIF3a, eIF3d, and SSU proteins S26, S28 (Pisarev, Kolupaeva, Yusupov, Hellen, & Pestova, 2008); and the possible location of the eIF4G-binding site for eIF4E in proximity to eIF3e (LeFebvre et al., 2006; Villa et al., 2013; Zhou et al., 2008), eIF3c, and eIF3d (Villa et al., 2013; Zhou et al., 2008). Approximately the same SSU location is designated for eIF3d, which can be considered a functional analogue of eIF4E for *JUN*-like mRNAs (Cate, 2017). The small size of eIF4E and the interaction with eIF4G through its dorsal surface (Figure 4e) also make the location of eIF4E close to the P-site-decoding center plausible, leaving little to no gap between the 5′ end and the 5′ UTR sequence inspected by the decoding center of the scanning SSU (Figure 6b).

The overall configuration of the early initiation complex in this case would include mRNA bound to eIF4E (in the form of eIF4F), eIF1A and eIF1 near the mRNA-binding cleft of the SSU, and eIF3 (possibly GTP:eIF2:Met-tRNA_i^{Met} as well). Upon SSU binding to the cap and accommodation of mRNA in the mRNA-binding cleft of the SSU, a structural rearrangement of the initial complex would then ensure that the mRNA becomes fully interlocked in the structure. The nature of the rearrangement and the trigger event are incompletely understood; however, detachment from the cap structure (see below) can be a coinciding signal, and adoption of eIF1A interactions that bridge SSU head and body, as well as complete accommodation of the TC, can form the initial mRNA “latch”. The proposed cooperative displacement of eIF3j from the A-site by eIF1, eIF1A, and mRNA (Sokabe & Fraser, 2014, 2017) and eIF3b binding near eIF2 at the intersubunit face of the SSU (Simonetti et al., 2016) can also be considered the signatures of the rearrangement.

An alternative model (more akin to the original “threading” proposal) that postulates eIF4E localization near the mRNA entry site of the SSU (at the solvent side closer to the A-site) was also suggested (Figure 6c) (Kumar et al., 2016). An entry-proximal eIF4E position is supported by the ability of the mammalian *in vitro* initiation system to recognize initiation codons very close to the mRNA 5′ end (1–2 nt) without apparent backscanning (Kumar et al., 2016). In this case, mRNA is supposed to be “extruded” into the preformed channel of the initiating SSU after the initial binding. Despite their differences, both models imply two distinct substeps of the process, initial mRNA-cap:eIF4F:initiating SSU interactions, that are subsequently rearranged into a scanning-competent interaction, with mRNA locked in the SSU independently on the cap structure. The position of the complete eIF4F, or even just the full-sized eIF4G in this complex, remains an important unanswered question; more precise localization of eIF4G would allow hypothesize on both the mechanism of mRNA loading on the SSU and the scanning motion (see below).

3.3 | Release of the cap from translating ribosomes

To allow for mRNA to be translated by multiple ribosomes (as observed by the formation of polysomes), SSU:cap interactions must be broken at some point during translation to enable a new SSU to bind at the 5′ end of mRNA. Alternative “cap-tethered” or “cap-severed” models were hypothesized for this process. In the cap-tethered model, the 5′ cap would remain in contact with the initiating SSU at least until the completion of the initiation. This model is compelling as no additional mechanisms are required to explain SSU detachment from the cap, which would simply be a consequence of the release of the initiation factors upon the assembly of elongating ribosomes. However, systematic increases of the 5′ UTR length of reporter mRNAs did not affect protein yield as measured in yeast cells (Berthelot, Muldoon, Rajkowitsch, Hughes, & McCarthy,

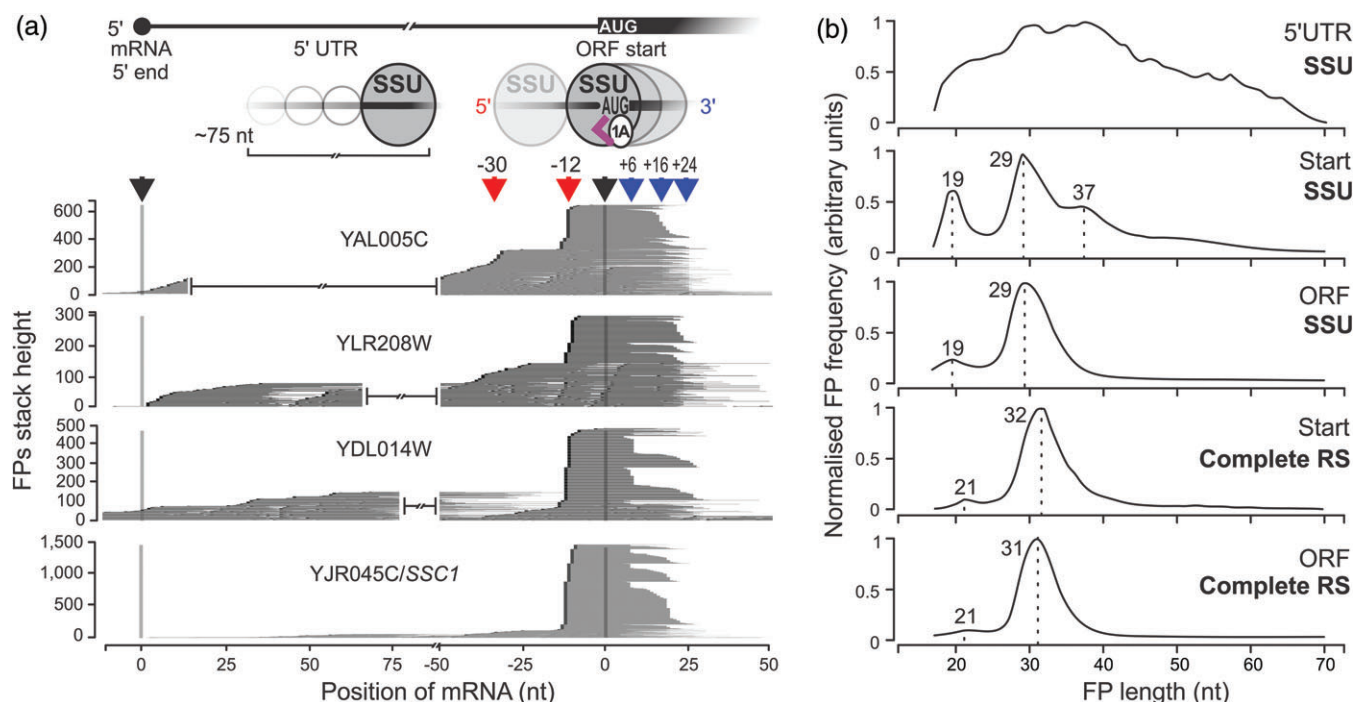


FIGURE 7 *In vivo* capture of translation initiation intermediates in yeast cells using translation complex profile sequencing (TCP-seq). (a) Footprints (FPs) of the SSUs fixed within 5' UTRs and at start codons of selected mRNAs. FPs are shown near the annotated mRNA 5' end ('0' position on left part of X-axis before break) or near the start codon in the region corresponding to ± 50 nt from the first nucleotide of the start codon ('0' position on right part of X-axis after break). Metagenic modal 5' (red) and three 3' (blue) FP extremities (corresponding to the stages of start codon recognition; see Figure 9 and the corresponding text) are marked. (b) FP length characteristics for SSU or complete ribosome (RS) complexes depending on position along mRNAs. TCP-seq data and some graphic elements were modified from (Archer et al., 2016)

2004). Provided that the availability of active eIF4E and "43S" complexes for attachment to mRNA caps was not a dominant limiting factor in these experiments, this would imply that multiple SSUs can simultaneously traverse 5' UTRs (Berthelot et al., 2004).

Early biochemical *in vitro* evidence using edeine, which renders scanning "infinite" by attenuating recognition of start codons, had also suggested that multiple SSUs can scan an mRNA simultaneously (Kozak & Shatkin, 1978). These experiments employed a feature of translation complexes, namely, that SSUs as well as full ribosomes protect their "footprint", the mRNA region they are attached to, from nuclease digestion (Barondes & Nirenberg, 1962; Jackson & Standart, 2015; J. A. Steitz, 1969). These ribosome footprints can be physically recovered and analyzed for footprint size and, potentially, location along mRNAs (Both, Furuichi, Muthukrishnan, & Shatkin, 1975; Jackson & Standart, 2015; Wolin & Walter, 1988; Kolupaeva, Pestova, & Hellen, 2000; Kozak & Shatkin, 1976, 1977). For example, this feature was used to discover pausing and stacking of elongating ribosomes, demonstrating that tightly packed disomes, trisomes *etc.* protect proportionally longer mRNA fragments (Wolin & Walter, 1988). SSU footprints were observed during translation initiation as well, although their positions could not be retrieved unambiguously with methods available at the time (Kolupaeva et al., 2000; Kozak & Shatkin, 1976, 1977). The combination of this approach with high-throughput sequencing gave rise to the highly successful "ribosome profiling" method (Baranov & Michel, 2016; Ingolia et al., 2009; Jackson & Standart, 2015), which was, at first, limited to recording ribosomes (stalled with antibiotics) during translation elongation. However, the recently developed translation complex profile sequencing (TCP-seq; Figure 7) method added a formaldehyde crosslinking step to the approach to preserve and detect footprint distribution of any of the translation complexes, including that of initiating SSUs (Archer et al., 2016; Shirokikh et al., 2017). TCP-seq showed no negative correlation between 5' UTR length and SSU start codon occupancy in live, exponentially growing yeast cells and no increased 5' end proximal accumulation of SSUs as might be expected from cap-tethered scanning (strong cap:eIF4E:eIF4G:SSU interactions resulting in lingering SSU dwell time at 5' ends). Moreover, a subset of SSU footprints representing late intermediates of start codon recognition were extended further upstream to -30 nt on mRNA, consistent with the presence of a second queued-up SSU (Figure 7a). These observations were made in meta-gene analyses of transcriptome-wide data as well as for a subset of individual mRNAs and are consistent with the concomitant traversal of 5' UTRs by multiple SSUs. These findings strongly suggest that scanning can take place in a cap-severed mode, although they do not formally exclude the coexistence of cap-tethered mode (Archer et al., 2016; Shirokikh et al., 2017).

Interestingly, recent biochemical observations demonstrated that breaking any link in the mRNA cap:eIF4E:eIF4G:eIF3 interactions, or absence of eIF4A, effectively turns eIF4E or eIF4E:eIF4G into a competitive inhibitor of initiation (Kumar et al., 2016). Even more intriguingly, eIF4E was absent from the cap during the later stages of initiation as determined by UV crosslinking (Kumar et al., 2016), although it is not certain whether eIF4E first detached from the cap or from eIF4G, with a subsequent decrease of affinity to the cap. Taking into account these data, it may be expected that eIF4E, eIF4A, eIF4G and eIF3 actively participate in the mRNA loading onto the initiating SSU, and the subsequent detachment of the initiating SSU from the cap occurs immediately upon the complete accommodation of mRNA (Figure 6b,c). This might involve some form of eIF4F disassembly, triggered by binding to the cap structure (Merrick, 2015). Similar to the mechanism known for prokaryotes, whereby the first elongation steps destabilize Shine–Dalgarno interactions (Uemura et al., 2007), cap detachment in this case may involve an active (energy-dependent) rearrangement coinciding with the commencement of the 3'-ward SSU motion. Due to the transient nature of the early scanning complexes, determining the exact sequence of events during cap-dependent mRNA attachment will require finely resolved, single-molecule kinetic evidence, such as for the eIF4E–cap interactions (O'Leary et al., 2013) and initiation on Cricket Paralysis Virus (CrPV) and Hepatitis C Virus (HCV) IRES structures (Fuchs et al., 2015; Johnson et al., 2017; Petrov, Grosely, Chen, O'Leary, & Puglisi, 2016).

4 | SCANNING

Scanning commences once the SSU has been loaded onto the mRNA 5' end. Key mechanistic aspects of scanning are the movement of SSUs along the mRNA and continuous inspection of mRNA bases for Watson–Crick interactions with the CAU anticodon of the Met-tRNA^{Met} (Figure 1a and middle part of Figure 2) (Kozak, 1978). Classic biochemical evidence indicates that the SSU motion is 5' → 3' directional rather than occurring merely by random bidirectional diffusion (Berthelot et al., 2004; Vassilenko, Alekhina, Dmitriev, Shatsky, & Spirin, 2011). However, a direct characterization of scanning intermediates has proven to be difficult due to the highly dynamic nature of the process. There are, thus, several intriguing open questions, such as what “drives” scanning and how is a net 3'-ward movement of the SSU achieved.

4.1 | Factor requirements for scanning

An operational distinction can be made between scanning as it occurs on completely linear (model) 5' UTRs and (natural) mRNA leaders that feature varying degrees of structure. The former can be traversed without energy expense, whereas ATP (or NTP) hydrolysis will be required to ensure progress through the latter. Accordingly, translation of mRNA 5' UTRs of poly(U), but not native sequences, can occur in *Artemia salina* embryos where many initiation factors are not active (Sierra, Meier, & Ochoa, 1974). All eIF4 group proteins were not required for initiation on a synthetic mRNA possessing an unstructured 5' UTR composed of (CAA) repeats; however, the other features typical for scanning, such as preferential recognition of the 5'-proximal start codons, were preserved (Pestova & Kolupaeva, 2002). The recognition of start codons is possible with a minimal set of eIFs 1, 1A, and 2 on mRNAs comprised of (U,C) sequences in yeast (Maag, Fekete, Gryczynski, & Lorsch, 2005) or featuring a poly(A) 5' UTR in a mammalian translation system (Shirokikh & Spirin, 2008). SSUs that remained attached to mRNA after termination were able to recognize nearby start codons in both directions (3' but also 5' through so-called “backward” or “retrograde” scanning) with the eIFs 1, 1A, 2 and 3 present. However, retrograde scanning was suppressed when eIFs 4F, 4A, and 4B were added (Abaeva, Pestova, & Hellen, 2016; Skabkin, Skabkina, Hellen, & Pestova, 2013). This suggests that SSUs can lock on mRNAs, remain attached, and slide randomly in the presence of eIFs 1, 1A, and 2 (and 3), while eIFs 4F or 4G, 4A, 4B, and ATP hydrolysis provide a 3' directionality to their motion. Other RNA helicases have been implicated to further assist scanning on highly structured mRNA leaders (Abaeva, Marintchev, Pisareva, Hellen, & Pestova, 2010; Dhote, Sweeney, Kim, Hellen, & Pestova, 2012; Parsyan et al., 2011; Pisareva, Pisarev, Komar, Hellen, & Pestova, 2008). Accordingly, while SSUs can navigate moderately structured 5' UTRs aided by eIFs 1, 1A, 2, 3, 4A, 4B, and 4F (Pestova & Kolupaeva, 2002), additional “support” in the form of DHX29 (Abaeva et al., 2010; Pisareva et al., 2008), DDX3X (Abaeva et al., 2010; Sen, Zhou, Ingolia, & Hinnebusch, 2015), or potentially DDX3 (Soto-Rifo et al., 2012), DHX9 (Ranji, Shkriabai, Kvaratskhelia, Musier-Forsyth, & Boris-Lawrie, 2011), DHX33 (Zhang, You, Wang, & Weber, 2015), and others, is required for more structured mRNAs.

More structured 5' UTRs have long been known to inhibit translation (Kozak, 1980), including the *in vivo* observation of inhibition by a ~ – 30 kcal/mol hairpin (Kozak, 1988). In a purified system with eIFs 4A, 4B, and 4F, hairpins with free energy from ~ – 20 kcal/mol appear to be inhibitory to translation initiation (Pisareva et al., 2008), even though the same factors in the absence of initiation complexes were able to destabilize much stronger duplexes (Ozes et al., 2011), of even up to ~ – 150 kcal/mol, by sequentially separating ~11 nt per step with substeps down to as low as 2 nt (Garcia-Garcia et al., 2015). Therefore, the activity of group 4 eIFs appears to be tightly modulated in the context of SSU scanning, such that their

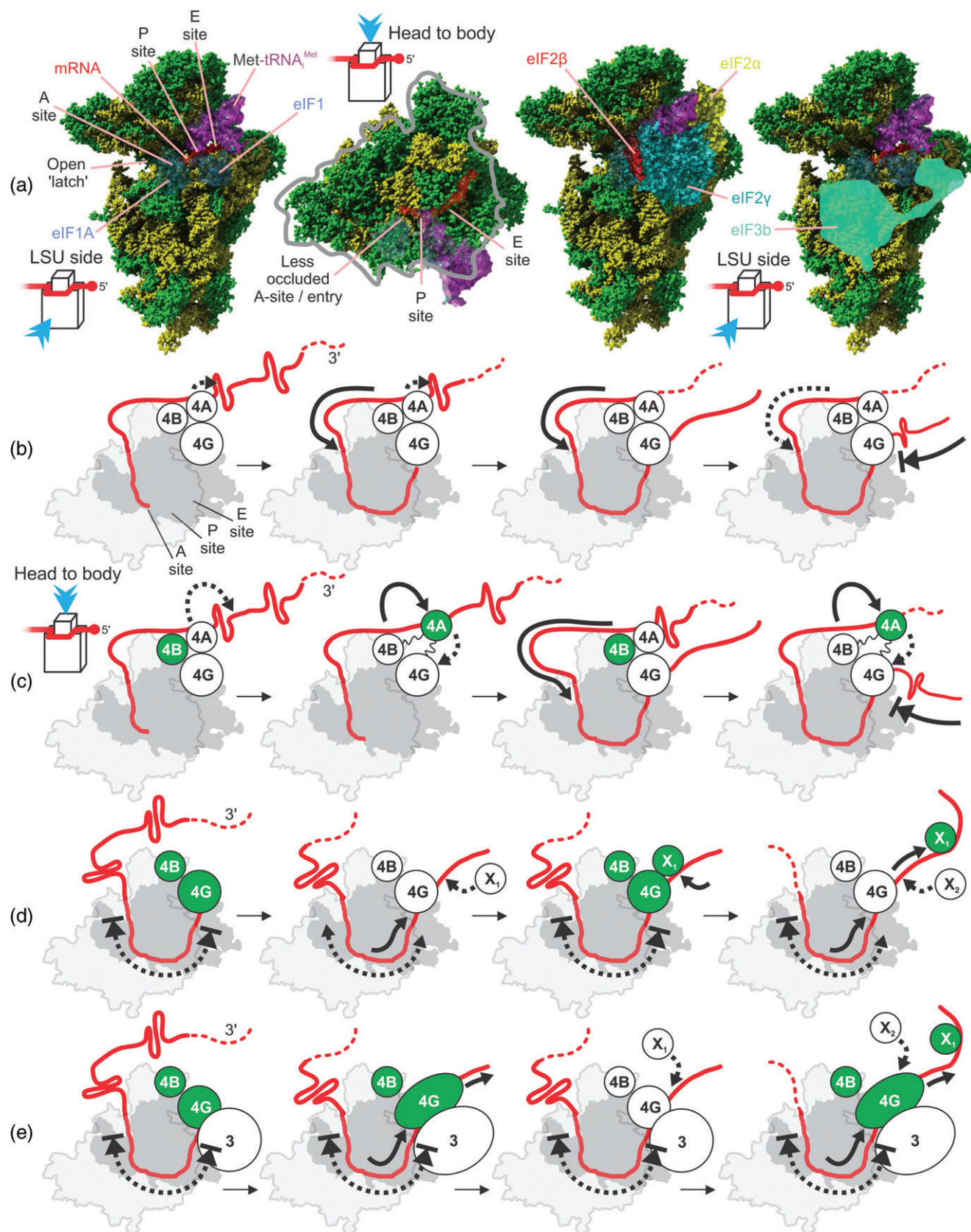


FIGURE 8 Legend on next page

mechanism of action may differ from that of the SSU-free form. For the better-studied example of DHX29, its relative low abundance (~10-fold lower than that of SSUs) (Pisareva et al., 2008) suggests an auxiliary role in the scanning, while its localization in the entry side of the mRNA-binding cleft (adjacent to the A-site) and strong direct interaction with SSUs (and not eIF4G) suggest a different mechanism compared to eIF4A (Abaeva et al., 2010; des Georges et al., 2015; Hashem et al., 2013a; Pisareva et al., 2008). However, given the high abundance of eIF4A in the cell and its known capacity to destabilize secondary structures through “mass action” by binding to the single-stranded RNA in its ATP-activated form, rather than by processive helicase activity (Aitken & Lorsch, 2012; Garcia-Garcia et al., 2015; Liu, Putnam, & Jankowsky, 2008), the prevalence and input of the additional RNA helicases into translation, as well as their potential allocation to specific mRNAs in live cells, remain to be fully elucidated.

4.2 | Structure of scanning complexes

Advances in the structural modeling of eukaryotic initiation complexes, combined with biochemical and genetic evidence acquired in the last decade, make it possible to outline several likely structural features of the scanning process (Figure 8a). Modeling suggests that SSUs adopt a configuration within scanning complexes that is distinct to factor-free SSUs and complexes involved in later steps of initiation or SSUs as part of the elongating ribosomes (Aylett & Ban, 2017; Cate, 2017; Erzberger et al., 2014; Hinnebusch, 2017; Voigts-Hoffmann, Klinge, & Ban, 2012).

In all SSU structures pertaining to “scanning”, if the resolution allows its modeling, eIF1 occupies the area immediately below the P-site and makes direct electrostatic contact with rRNA (Erzberger et al., 2014; Hussain et al., 2014; Rabl, Leibundgut, Ataide, Haag, & Ban, 2011; Weisser, Voigts-Hoffmann, Rabl, Leibundgut, & Ban, 2013). This “insertion” of eIF1 between tRNA and SSU is supportive of its main function in promoting the fidelity of start codon selection by destabilization of codon–anticodon interactions (Hussain et al., 2014; Lomakin & Steitz, 2013; Rabl et al., 2011; Weisser et al., 2013). The central, well-structured part of eIF1A, another small intersubunit side-bound factor, bridges the SSU head and body directly over the A-site and encloses the mRNA in the mRNA-binding cleft of the SSU, a feature deemed responsible for the processivity of scanning motion. eIF1A localization at the A-site is also hypothesized to prevent tRNAs from binding to the SSU A-site, a role also performed by its bacterial counterpart IF1 (Lomakin & Steitz, 2013; Weisser et al., 2013). The N-terminal portion of eIF1A is distinctively less structured in the “scanning” complexes (Lomakin & Steitz, 2013; Weisser et al., 2013). eIFs 1 and 1A are known to bind to the SSU cooperatively, but this is commonly explained by their local effects on the nearby SSU rRNA (affecting mostly helix 44) and possibly by interactions with the other factors, rather than their direct binding to each other (Maag & Lorsch, 2003; Weisser et al., 2013).

eIF2 α is located at the E-site in the scanning complexes but is less deeply inserted into the mRNA-binding cleft of the SSU compared to the other initiation complexes. eIF2 α does not make contact with the –3 nt position of mRNA, which is a critical component of the consensus start codon nucleotide context (see below in the “Recognition of start codons and joining

FIGURE 8 Models of scanning SSU configuration and possible determinants of scanning motion directionality. (a) Combination of the atomic cryo-EM reconstruction structure (PDB entry 3JAO) and cartoon schematics of the scanning SSU with the scanning-conducive, relatively “open” mRNA latch and widened and less-obstructed mRNA entry marked. Met-tRNA^{Met} is in the P_{OUT} conformation, leaning toward the E-site and with a loose and more lateral positioning in the P-site. Surfaces of only eIF1, eIF1A (left), and these factors together with eIF2 or eIF3b (right) are shown for simplicity. Atomic ball-and-stick model of rRNA is shown in yellow and ribosomal proteins in green. Gray line on the head-to-body view outlines the SSU head. The mRNA position was taken from the “closed” SSU structure (PDB entry 3JAP) aligned with 3JAO by the rRNA trace in the SSU body region. The superimposed eIF3b cartoon was taken from Figure 3g. (b)–(e) Cartoons illustrating different concepts and models to explain 5' → 3' directional motion during scanning. SSU schematics and indications as in Figure 3. Possible alterations of states of higher (green) and lower (blank) factor affinity to mRNA are shown where applicable. (b) eIF4 group factors perform mRNA “unwinding” in front of the SSU as the structure of the initiation complex directs them to the incoming portion of mRNA (SSU entry) (left plot). The 3'-ward stretch of single-stranded mRNA then allows the SSU to spontaneously relocate (diffuse) onto it (middle two plots). Secondary structure can refold upon the mRNA exiting SSU and ensure motion directionality as no helicase activity is present at the SSU exit site (right plot). (c) eIF4 group factors perform “unwinding” in front of the SSU by cycles of clamping to the single-stranded mRNA. eIF4B has a higher affinity to mRNA (green) while in close contact with eIF4A (left). Upon eIF4A conformational change, eIF4A rebinds to the downstream (incoming) portion of mRNA (green; middle left plot) and then restores its tight binding to the complex, bringing the downstream portion of mRNA along into the SSU and stimulating eIF4B's high affinity to mRNA (green; middle right plot). The cycle then repeats (right plot). Note that (b) and (c) are most compatible with an entry-side “threaded” mRNA loading at the beginning of initiation. (d) eIF4 group factors are sequentially lodged behind the SSU as it takes small 3'-ward advances due to the random (thermal) motion and act as anti-rollback “pawl” structures. All factors are tightly bound to the SSU, mRNA, and to each other (left plot; green). Due to an induced conformational change, mRNA-bound factor(s) lose(s) affinity to the rest of the complex but not to mRNA. This allows 3'-ward SSU sliding along mRNA, with accidental and sequential unwinding of some of the downstream structure (middle left plot). When a sufficient exit portion of the single-stranded mRNA is exposed, a new molecule of mRNA-binding factor (X₁) rebinds it, preventing backward SSU sliding (middle right plot). The cycle repeats with the next RNA-binding molecule (X₂; right plot). (e) Similar to (d), but eIF4-group factors (eIF4G) and eIF3 also undergo a large-scale conformational change, possibly induced by the eIF4A enzymatic cycle, and actively pull out a portion of mRNA 3'-ward, after which they rebound mRNA more proximally to the SSU. Note that (d) and (e) are most compatible with a lateral loading (slotting) of mRNA into the SSU mRNA-binding cleft, where cap:SSU interactions begin near mRNA exit site

by the large ribosomal subunit" section). eIF2 β and eIF2 γ do not make much contact with the SSU and mostly extend into the solution at the intersubunit side (Hashem et al., 2013a; Hussain et al., 2014). Importantly, Met-tRNA_i^{Met}, bound to eIF2, adopts a scanning-specific conformation where it is reaching less deeply in the P-site overall; its anticodon is laterally more distant from the P-site mRNA (by ~ 7 Å), contributing to a wider configuration of the P-site, compared to the other initiation complexes (Hussain et al., 2014; Llacer et al., 2015). While the Met-tRNA_i^{Met} maintains its main contacts with the SSU head, head motion (see below) bends the anticodon stem loop of the Met-tRNA_i^{Met} and laterally distances it from the SSU body (Llacer et al., 2015). Deeper positioning of Met-tRNA_i^{Met} into the P-site is prevented by eIF1 at this stage.

Consistent with early findings (Emanuilov, Sabatini, Lake, & Freienstein, 1978; Siridechadilok et al., 2005; Srivastava, Verschoor, & Frank, 1992), the "core" of eIF3 assembled from PCI/MPN domains (eIF3 a, c subunits in yeast; a, c, f, h, e, k, l, m in mammals) resides on the solvent side of the SSU, closer to the exit side of the mRNA-binding cleft and between the head and body of the SSU. A second volume of compact density can be attributed to the eIF3b, eIF3g, eIF3i, and eIF3j subunits localized on the solvent and LSU sides of the SSU, but not near the entry side of the mRNA-binding cleft (Aylett, Boehringer, Erzberger, Schaefer, & Ban, 2015; des Georges et al., 2015; Erzberger et al., 2014; Hashem et al., 2013a; Llacer et al., 2015; Simonetti et al., 2016). Thus, eIF3 remarkably engulfs SSU from both the solvent and LSU sides, with the extensions of its "core" a, c subunits and mRNA-entry b, g, i assembly reaching eIFs 2 and 5 at the mRNA exit side and 1A at the entry side, respectively (see Figures 3f,g) (Llacer et al., 2015; Simonetti et al., 2016). eIF3a and eIF3b subunits are thought to provide a relatively thin and flexible link between these eIF3 substructures.

Recent studies show much apparent movement of many eIF3 domains and subassemblies across the SSU in synchronization with different initiation stages. For example, cryo-EM reconstructions of near-native "scanning" complexes derived from nonfractionated lysate did not reveal eIF3j density in the entry side of the mRNA-binding cleft (Simonetti et al., 2016), unlike other reconstructions where no mRNA was present (Aylett et al., 2015). These data are consistent with the hypothesis derived from biochemical experiments that eIF3j can function to antagonize the binding of mRNA and can thus become lost from the complexes altogether or detached from the SSU mRNA-binding cleft early on during initiation (Sokabe & Fraser, 2014), as confirmed by its absence in the initiation complexes reconstituted at start codons (Unbehaun et al., 2004). Furthermore, relocation of eIF3b to the LSU side near the mRNA entry and below the mRNA-binding cleft, where it may interact with eIF2 γ and additionally stabilize the position of eIF2 during scanning, has been reported as one of the large-scale rearrangements (Figure 3g) (Simonetti et al., 2016). Another notable difference to the SSUs with out mRNA is the tilting of the whole eIF3 "core" PCI/MPN assembly down from the SSU head (Simonetti et al., 2016).

At least some of the mammalian scanning complexes may additionally contain DHX29 interacting with eIF3 subunits b and g directly in the entry site of the mRNA-binding cleft (Figure 5c) (des Georges et al., 2015). There may be other constituent parts to the scanning complexes that were either not included in the *in vitro* complex preparations or could not be modeled due to insufficient resolution. For example, in some cryo-EM reconstructions, electron density contacting eIF2 α , eIF2 β , eIF1, and possibly Met-tRNA_i^{Met} can be attributed to eIF5 (Hussain et al., 2014), but a more definitively resolved position of this factor is currently not available.

The overall mRNA path in the scanning complexes may be similar to the other ribosome-mRNA structures, although it lacks the pronounced kink between A- and P-sites characteristic of ribosomes with A- and P-site paired tRNAs or bacterial translation complexes. It is widely accepted that this smoother passage may facilitate SSU sliding along the mRNA, which is a requirement for scanning (Hussain et al., 2014; Llacer et al., 2015; Lomakin & Steitz, 2013). Interestingly, mRNA wraps around at least half of the SSU between head and body, with the inbound and outbound "trajectories" of mRNA so close together they could overlap at the solvent side (Hussain et al., 2014; Llacer et al., 2015). The mRNA is less constrained at the entry side relative to the exit side, where the mRNA-binding cleft is occluded by the densities of eIF1A in the A-site and further 5'-ward by Met-tRNA_i^{Met} and eIF1 at the P-site (Hussain et al., 2014; Llacer et al., 2015). The "head" of the SSU appears to be rotated clockwise by about 3–8°, placing the head P-site slightly toward the body A-site positions compared to the "empty" SSUs, which is hypothesized to aid in avoiding a steric clash with the Met-tRNA_i^{Met} anticodon (Hussain et al., 2014; Llacer et al., 2015; Lomakin & Steitz, 2013). Cryo-EM reconstructions, some of which included a model mRNA, also suggest a pronounced movement of the SSU head further away from the "body", toward the solvent side of the SSU (Llacer et al., 2015; Passmore et al., 2007). Overall, the configuration of the SSU described above is most commonly referred to as "open, scanning-competent" or "P_{OUT}" (suggesting a more peripheral, outer localization of Met-tRNA_i^{Met} relative to the SSU P-site). Some cryo-EM reconstructions notably feature the open "latch" near the entry side of the mRNA-binding cleft of the SSU, which is formed by rRNA helix 34 of the head and helix 18 of the body of the SSU (Figure 8a) (Llacer et al., 2015; Passmore et al., 2007).

Many features observed in the structures of the putative scanning configuration are consistent with the available cross-linking, protection, and mutagenesis data, and hence, a more complete picture can be drawn by merging the information from all sources. For example, eIFs 1 and 1A have been previously mapped by the directed hydroxyl radical cleavage method to

their respective localizations at the P and A-sites (Lomakin, Kolupaeva, Marintchev, Wagner, & Pestova, 2003; Yu et al., 2009). The role of eIF1 in destabilizing codon–anticodon interactions during scanning has been thoroughly investigated (Lomakin, Shirokikh, Yusupov, Hellen, & Pestova, 2006; Martin-Marcos et al., 2014; Nanda, Saini, Munoz, Hinnebusch, & Lorsch, 2013; Pestova, Borukhov, & Hellen, 1998; Zhang, Saini, Shin, Nanda, & Hinnebusch, 2015). The eIF1A N-terminal and C-terminal extensions (unstructured in the “free” eIF1A form) (Zhang et al., 2015) have been shown to stabilize codon–anticodon interactions and the “open”, scanning-competent configuration of the SSU and TC, respectively (Cheung et al., 2007; Fekete et al., 2007; Nanda et al., 2013; Saini, Nanda, Lorsch, & Hinnebusch, 2010). Conversely, the eIF1A N-terminal portion is delocalized in the scanning configuration, while the C-terminal extension is involved in the interactions at the SSU P-site (Zhang et al., 2015). eIF3c has been implicated in binding to eIF1 and eIF5 in the MFC and possibly in “scanning” complexes (Asano, Clayton, Shalev, & Hinnebusch, 2000; Fletcher, Pestova, Hellen, & Wagner, 1999; Luna et al., 2012; Phan et al., 1998; L. Valasek, Nielsen, Zhang, Fekete, & Hinnebusch, 2004), with its N-terminal extension reaching toward the P-site (Erzberger et al., 2014; Obayashi et al., 2017). Indeed, eIF3c (likely through its interactions with eIF1 and eIF5) is known to affect the fidelity of the start codon selection (Karaskova et al., 2012; L. Valasek et al., 2004). Nuclear magnetic resonance (NMR) chemical shift perturbation suggests that the N-terminal extension of eIF3c partially overlaps with eIF1 interaction surfaces on the SSU and thus may moderate eIF1 function to destabilize codon–anticodon interactions (Obayashi et al., 2017). These findings, combined with mutagenesis and analytical ultracentrifugation data, suggest that the C-terminal portion of eIF5 may bind the N-terminal part of eIF1A in the scanning complexes, resulting in increased affinity of the complex to eIF1 *via* eIF5:eIF1 interactions and in decreased binding of the C-terminal part of eIF5 to the N-terminal part of eIF2 β . This lack of tight eIF2:eIF5 interaction may cause inhibition of GTP hydrolysis in the TC during scanning, which would prevent premature fixation on poorly recognized triplets (Luna et al., 2012; Nanda et al., 2013; Obayashi et al., 2017). Thus, while eIF1A is one of the most conserved initiation factors and is a direct homologue of prokaryotic IF1, it appears to have acquired eukaryote-specific roles in coordinating scanning through its C- and N-terminal extensions. The factor is therefore remarkably persistent during the whole process of initiation, linking eukaryotic scanning with the later stages of initiation that are well-conserved across all life.

In vitro reconstitutions of the “scanning” complexes and their probing with directed hydroxyl radicals produced at different locations of eIF1A demonstrated substantially more accessible rRNA in the A-site and P-site localized helices (Zhang et al., 2015). rRNA residues located more distantly to the A- and P-sites near the entry side of the mRNA-binding cleft also displayed elevated accessibility, including helices 18 and 34 that form the mRNA “latch” of the SSU (Zhang et al., 2015). Importantly, the observed less-tight interactions of mRNA with SSUs in the reconstituted *in vitro* models of scanning complexes are recapitulated *in vivo* in the form of shorter nuclease protection observed on the entry side of the scanning SSUs, compared to the later stages of initiation and to the protection of elongating ribosomes with TCP-seq (Figure 7) (Archer et al., 2016). The minimal protection length of ~19 nt (Archer et al., 2016) appears overall concordant with, but somewhat lower than, the 22 nt of the cryo-EM-resolved mRNA (Hussain et al., 2014), suggesting that some parts of the SSU-bound, less-flexible mRNA are still accessible to the nuclease digestion. While this cannot be used to rule out the “closed latch” configuration of the scanning SSUs, it strongly evidences higher accessibility of the mRNA in the entry channel of the SSU and lack of stable protection of mRNA in the entry side of the mRNA-binding cleft on average *in vivo*.

Overall, there are multiple indications that the structure of scanning complexes can be highly dynamic. The observed uncertainty of the entry channel configuration may, in part, be a consequence of the differences in the methods of preparation and in the overall composition of the reconstituted complexes. However, we may also expect large-scale functional molecular motions that depend on the energy expense during scanning, such as those proposed for alterations between the “open” and “closed” states of the SSU induced by corresponding “closed” ATP- and RNA-bound and “open” eIF4A conformations (Sokabe & Fraser, 2017). In this case, different *in vitro* reconstitution models may artificially stabilize a particular conformation of the complexes (as often occurs, *e.g.*, in crystallographic studies), resulting in a somewhat different overall geometry. Most importantly, even the combined structural and biochemical data do not unanimously outline the position and possible orientation of eIF4F or eIF4G in any of the initiation complexes, including those related to scanning. Of all eIF4 group factors, only the position of eIF4B was determined in yeast to be at the “head” on the solvent side of the SSU, closer to the exit side of the mRNA-binding cleft (Walker et al., 2013). These data and those on eIF4G interactions with eIF3 subunits c, d, and e allow us to hypothesize on the position of eIF4G (LeFebvre et al., 2006; Villa et al., 2013). Because eIF4B can interact with eIF4A within eIF4F, and the main eIF4A- and eIF3-binding sites are relatively close together in eIF4G (Nielsen et al., 2011), it may be expected that the central part of eIF4G colocalizes near the eIF4B-binding site on the SSU. Even more speculative is the orientation of eIF4G on the scanning SSU given that approximately half of the full-size eIF4G has a highly elongated geometry and extends for ~200 Å, capable of connecting virtually any two points on each SSU side (Nielsen et al., 2011). Based on the closer location of the eIF3c- and eIF3d-interacting portions of eIF4G to its N-terminus, compared to the more C-terminal location of eIF3e-binding site (Villa et al., 2013), the overall linear geometry of the eIF4G middle

domain (Nielsen et al., 2011), and the mammalian eIF3 orientation on the SSU, the middle portion of eIF4G may be located with its N-terminus closer to the head on the back of the SSU, and the C-terminal part reaching to contact eIF3e down at the body of the SSU (Figure 4d, right). However, other possibilities have also been outlined (Kumar et al., 2016), leaving the orientation of this factor on the SSU a major open question. Knowing the positions of the eIF4 group factors will be critical to discern the exact molecular mechanism of the $5' \rightarrow 3'$ motion during scanning.

4.3 | Possible determinants of scanning directionality

To explain the $3'$ -ward SSU motion on mRNA, two main concepts are currently known, which differ in the likely roles and localizations of the main eIF4 group factors in the scanning SSUs. The first concept has eIF4A “unwinding” stable RNA helices in front of the scanning SSU (in cooperation with eIF4G and eIF4B/eIF4H and possibly another RNA helicase or single-stranded RNA-binding protein) (Figure 8b) (Marintchev, 2013; Marintchev et al., 2009; Marintchev & Wagner, 2004; Parsyan et al., 2011). These “unwinding” factors would thus localize at or before the leading edge of the SSU, which may be called “SSU-trailing” or “pulled SSU” scanning. However, while the unwinding of the secondary structures of mRNA may indeed be required for the $3'$ -ward motion of the SSU over mRNA (given that only single-stranded RNA can be efficiently scanned), this model does not provide systematic answers for the directional, powered motion, likely conferred by the eIF4 group factors (Fraser, 2015; Garcia-Garcia et al., 2015; Kumar et al., 2016; Ozes et al., 2011; Skabkin et al., 2013; Zinoviev, Hellen, & Pestova, 2015). The reassembly of RNA structure behind scanning complexes could explain directionality. However, one cannot expect an even distribution of secondary structure or RNA–protein interactions across the entire length of $5'$ UTRs, and backscanning is not readily observed on unstructured mRNAs in the presence of eIF4F, eIF4A, and ATP (Abaeva et al., 2016; Kumar et al., 2016; Skabkin et al., 2013). More experiments measuring the dynamics of scanning will be required to determine if directional scanning is efficient on unstructured mRNA. Alternatively, still with the scanning “motor” leading the SSU, directionality could be conferred by the eIF4 group factors continuously rebinding more $3'$ -proximal single-stranded portions of mRNA (and consequently “unwinding” or stabilizing them in the single-stranded form) while they bind or remain attached to the SSU, with mRNA “fed” into the mRNA-binding cleft of the SSU (Figure 8c). This mechanism would thus resemble the well-studied kinesin locomotion along microtubules (Kull & Endow, 2013; Rice et al., 1999). In this case, eIF4 group factors would serve not only as the $3'$ -directed RNA helicases but also as the ratchet-and-pawl structure preventing thermal rollback of the scanning SSU, a function postulated based on the known alterations of eIF4A affinity to mRNA in the ATP- and ADP-bound or the cofactor-free states. The observed intrinsic preference of eIF4A in complex with eIF4G to the $3'$ -portions of single-stranded oligonucleotides may compliment the mechanism of directionality in this case (Ozes et al., 2011), whereas the reciprocal alterations of high and low affinity of the SSU and eIF4A to mRNA (Sokabe & Fraser, 2017) may explain a kinesin-like step-wise directional advance. However, the appeal of this model is lessened given that the persistence length of single-stranded polyribonucleotides falls in the range of 3–4 nt (H. Chen et al., 2012; Chi, Wang, & Jiang, 2013). Thus, mRNA would need to be tightly “channeled” in the scanning complex to support the powered motion and not bulge out of the mRNA-binding cleft of the SSU and between it and the helicase site of action, an assumption that is not readily compatible with the more “open” entry-side configuration of the *in vivo* and *in vitro* SSUs and more tight interactions with mRNA in the P-site and near the mRNA exit.

The second concept involves the primary action of eIF4A (or its functional substitute) while located behind the SSU relative to mRNA, in what can be called an “SSU-leading” or “pushed SSU” scanning. In this case, the directionality of motion would be ensured by eIF4A or other associated factors (such as eIF4B or eIF4H), by strongly binding to upstream mRNA portions exposed by random (thermal) short-distance SSU movements along mRNA, thus preventing backward SSU motion in the ATP-bound state (Nielsen et al., 2011; Siridechadilok et al., 2005; Spirin, 2009a). Rebinding of the new “motor” factors in the gap of single-stranded RNA upon spontaneous $3'$ -ward relocation of SSU would thus serve as the “pawl” and result in the sustained, directional locomotion (Figure 8d). Thus, the SSU in this case is an integral part of a complex RNA helicase machine, consistent with the proposed function of ribosome as a processive RNA helicase in prokaryotes (Qu et al., 2011; Takyar, Hickerson, & Noller, 2005). The lodgement of the factors on mRNA behind SSU in this case may additionally provide the necessary “power stroke” of the locomotion by actively displacing SSU $3'$ -ward (Figure 8e). This latter scenario is similar to the translocation of the complete elongating ribosome or the addition of the newly synthesized residues during DNA replication coupled with the activity of processive DNA helicases, which also use an antirollback mechanism to select for directionality of motion (Kuriyan & O'Donnell, 1993).

The notable structural resemblance of mammalian eIF3 and its direct homology with the proteasome lid and Constitutive photomorphogenesis (COP) 9 signalosome are often discussed (Cate, 2017; Pick, Hofmann, & Glickman, 2009) and raise questions as to why such a complex structure was duplicated during evolution. What is perhaps less appreciated is the remarkable mechanistic similarities of the processes in which eIF3 and proteasome lid are involved. In the eIF3-like proteasome, COP9, Initiation factor 3 (PCI)/Mpr1/Pad1 N-terminal (MPN) core regulates the recognition of the ubiquitylated

portion of the protein, its subsequent irreversible binding (“commitment”) to the proteasome, and removal of the ubiquitinylation once the first unfolded portion of the protein enters the proteasome core (Matyskiela, Lander, & Martin, 2013). The PCI/MPN lid, in complex with α -helical ring-shaped AAA-ATPases, is thought to provide energy for the unfolding of the protein by stronger binding of the unfolded chains in the ATP-bound state and then to participate in the sequential translocation of the unfolded peptide into proteasome (Matyskiela et al., 2013). The whole sequence of events is reminiscent of mRNA cap recognition, detachment from the cap, and the subsequent ATP-dependent pulling of mRNA. The most actively mobile proteasome subunit is Rpn11 located in the center of the PCI domain assembly (Matyskiela et al., 2013) and is homologous to the eIF3f (Cate, 2017). The absence of eIF3f in yeast may be functionally compensated for by the eIF3a PCI domain portion that occupies a similar location and is proposed to substitute for the possible SSU-interacting functions (Aylett et al., 2015). Thus, it is tempting to speculate on a more active involvement of eIF4G and eIF3 PCI/MPN core in the large-scale motions that may be induced in these factors by eIF4A cycling through its “closed” and “open” states while it is bound to eIF4G (Figure 8e). Overall, the pushed-SSU model and the proteasome lid homology may be more supportive for the eIF4G localization on the solvent side of the SSU and near the mRNA exit (with the abovementioned caveat that its elongated shape allows it to reach between any points on one side of the SSU). eIF4G interactions with eIFs 4A, 4B, and 4H, and relatively high abundance of these factors, further suggest the possibility of their transient binding to mRNA in multiple copies behind the scanning SSU. In this case, possible dissociation and reassociation cycles of eIF4F components driven by eIF4A-mediated ATP hydrolysis, as suggested in multiple studies based on the biochemical properties of these factors (Gingras, Raught, & Sonenberg, 1999; Merrick, 2015; Pause, Methot, Svitkin, Merrick, & Sonenberg, 1994; Yoder-Hill, Pause, Sonenberg, & Merrick, 1993), may explain controlled recruitment and release of the new “pawl” factors behind the SSUs. However, the ratchet-and-pawl concept is not completely incompatible with the entry-side localization of eIF4G-attracted eIF4A on mRNA either. Moreover, eIF4 group factors may, in fact, act on both incoming and outbound portions of the scanned mRNA, where the structure of the incoming mRNA is destabilized by the additional helicases such as DHX29 or DDX3 (Soto-Rifo et al., 2012), and also eIF4A, which is later transferred to the outbound portion of mRNA.

Interestingly, *in vivo* TCP-seq data show substantial heterogeneity of footprint sizes for scanning SSUs (ranging from ~17 to ~70 nt). The additional regions of protected mRNA are on the 5' side of the SSU (Figure 7) (Archer et al., 2016). This “trailing tail” behind the scanning SSUs may be a result of the proximity to some large additional structures in the scanning complexes, such as eIF3 and eIF4G. The overall span of eIF4G can potentially cover single-stranded polyribonucleotide of a comparable length, but the extensions of 5' UTR to ~30 nt were reported to be sufficient to stabilize eIF3 binding (Unbehaun et al., 2004). However, even if the outbound mRNA is making tight contact with these structures, it is difficult to explain the relatively homogeneous distribution of the protected fragment lengths over the entire range (Figure 7b) and the fast disappearance of this protection once SSUs arrive at start codons (Archer et al., 2016). The additional 5'-ward protection more likely results from transient, multiple, but small mRNA-binding units, such as eIF4A or eIF4B, as outlined above. Defining the exact nature of the extended mRNA protection generated by the scanning SSUs thus appears critical to the understanding of the mechanism of scanning motion. Overall, the debate over what provides scanning directionality can be seen as part of a deeper issue. As was already emphasized in the literature, there is no absolute certainty of the mechanism of action for even the well-studied, “processive” replicative helicases. Both, the active duplex separation and the advantageous binding to single-stranded patches stripped by thermal motion are probable (Delagoutte & von Hippel, 2001; Phelps, Lee, Jose, von Hippel, & Marcus, 2013; Takyar et al., 2005; von Hippel & Delagoutte, 2001). Solving the mechanism of scanning motion presents a unique opportunity to address this question for a larger macromolecular structure where the localization of the helicase can be discerned.

The questions raised above can be expected to be even more intriguing to solve for translation initiation in higher eukaryotes. Indeed, it is well-known that yeast 5' UTRs are short and somewhat less structured compared to, *e.g.*, mammalian 5' UTRs (Tuller, Rupp, & Kupiec, 2009). Furthermore, the eIFs constituting the mammalian scanning machinery are more divergent compared to yeast. In *S. cerevisiae* (for which TCP-seq data were obtained), eIF3 has a substantially reduced number of core subunits, and its interaction with the SSU can be somewhat different. *S. cerevisiae* eIF4G lacks binding sites for eIF3, MNK, eIF2 β , and the second eIF4A binding site and, instead, interacts with SSUs directly. *S. cerevisiae* do not encode eIF4H, and their eIF4B does not readily dimerize, unlike in mammals or higher plants (Cheng, Sultana, Goss, & Gallie, 2008). There is a considerably smaller overall variation of eIF isoforms and alternatives in yeast, especially so for the group 4 eIFs and translation-associated RNA helicases. Thus, initiation in higher eukaryotes promises elevated complexity and also a “more complete” scanning machinery, which would be extremely interesting to investigate and which may more explicitly exhibit features of the scanning “motor”. We can expect to encounter even more scanning roadblock events due to the more structured, long, and complex 5' UTRs and a more diverse set of active RBPs, which creates a possibility to directly detect pronounced internal ribosome entry (especially in the longer 5' UTRs) and better understand the dynamics of the process (Shirokikh et al., 2017). At the same time, while there seems to be no technical limitations to perform TCP-seq in cells other

than yeast, the attribution of SSU footprints to mRNA isoforms, especially those with variations in the 5' UTRs, may present a problem (Shirokikh et al., 2017). Developing algorithms for isoform deconvolution could become useful to solve these problems (Kanitz et al., 2015; Martinez-Nunez & Sanford, 2016; Wang, McManus, & Kingsford, 2016).

4.4 | Noncanonical scanning and alternative models

Some examples of noncanonical scanning have been documented, and alternative mechanisms for inspection of 5' UTRs by the SSU have also been suggested (Haimov, Sinvani, & Dikstein, 2015). The experimentally tested cases involve discontinuous SSU migration along the 5' UTR, or "shunting", a mechanism that was originally proposed to explain the apparent skipping of a large portion of the Cauliflower Mosaic Virus (CMV) 35S RNA by the scanning SSUs (Futterer, Kiss-Laszlo, & Hohn, 1993). Shunting was suggested to be involved in many other cases of initiation on viral RNAs ("tripartite" 5' UTR of late adenoviral RNA (Yueh & Schneider, 1996); RNA of plant pararetroviruses (Hemmings-Mieszczak & Hohn, 1999; Pooggin, Ryabova, He, Futterer, & Hohn, 2006); Sendai virus Y proteins mRNA (de Breyne, Simonet, Pelet, & Curran, 2003); avian reovirus S1 mRNA (Sherrill & Lloyd, 2008)), but also on certain cellular sequences (Heat shock 70 protein mRNA (Yueh & Schneider, 2000); cellular Inhibitor of Apoptosis 2 (cIAP2) mRNA (Sherrill & Lloyd, 2008); and possibly others (Koh, Edelman, & Mauro, 2013; Lammich, Schobel, Zimmer, Lichtenthaler, & Haass, 2004; Mihailovich, Thermann, Grohova, Hentze, & Zacchetti, 2007; Rogers Jr., Edelman, & Mauro, 2004).

The exact molecular mechanism of shunting remains unidentified; however, it was proposed that special hairpin structures and surrounding sequences not only impede continued linear motion by scanning SSUs but also cause them to relocate to regions downstream (Dominguez et al., 1998; Hemmings-Mieszczak & Hohn, 1999; Yueh & Schneider, 1996). In addition, uORFs often found in the shunted 5' UTRs were demonstrated to facilitate shunting compared to continuous scanning, suggesting that reinitiation may be a requirement for nonlinear SSU movement (Pooggin, Futterer, & Hohn, 2008; Pooggin, Hohn, & Futterer, 2000). In this case, scanning would be resumed by SSUs that remain attached to the mRNA, similar to the well-studied examples of reinitiation upon termination on cellular (Gunisova et al., 2016; Gunisova & Valasek, 2014; Alan G Hinnebusch, 2005; Vattam & Wek, 2004; Zhang & Hinnebusch, 2011) or viral (Luttermann & Meyers, 2007, 2014; McCormick, Salim, Lambden, & Clarke, 2008; Meyers, 2003; Napthine et al., 2009; Powell et al., 2011; Poyry & Jackson, 2011; Poyry, Kaminski, Connell, Fraser, & Jackson, 2007; Poyry, Kaminski, & Jackson, 2004; Somers, Poyry, & Willis, 2013) uORFs, but ahead of a stable structure (hairpin). Thus, shunting may require rebinding of the scanning SSUs downstream across a stable structure while they remain attached to the initial location on the mRNA (Hemmings-Mieszczak & Hohn, 1999; Hemmings-Mieszczak, Hohn, & Preiss, 2000; Pooggin et al., 2008). *In vitro* reconstructions provide direct evidence that, in certain conditions, canonical factors responsible for scanning can assist in the bypassing of stable stem loops. Passing of stable structures through the entry side, but not the exit side, of the SSU mRNA-binding cleft was proposed to occur in the presence of eIFs 1, 1A, 2, 4A, 4B, and 4G (Abaeva et al., 2010), suggesting that efficient skipping of 5' UTR portions during shunting may require additional special interactions with mRNA and/or an incomplete (*e.g.*, post-termination) scanning SSU configuration.

Activities of DENR:MCT-1 and the homologous eIF2D (ligatin) (Figure 5a,b) were demonstrated during reinitiation (Jackson, Hellen, & Pestova, 2012; Skabkin et al., 2013; Zinoviev et al., 2015). They can explain cases of eIF2-independent initiation (Akulich et al., 2016; Dmitriev et al., 2010; Skabkin et al., 2010), which were reported to be responsible for the specific translation of uORF-containing mRNAs promoting proliferation and oncogenic regulation, as well as genomic instability (Hsu, Shi, & Gartenhaus, 2005; Kasiappan et al., 2009; Prosnia et al., 1998; Schleich et al., 2014; Tseng et al., 2017). Interestingly, while the mRNA "latch" remained closed in the recent structural reconstructions with these factors, the mRNA-binding cleft of the SSUs appeared to be less occluded, mostly due to the absence of eIF1A and eIF2 α (Lomakin et al., 2017; Weissner et al., 2017). eIFs 1 and 1A were also shown to antagonize activity of eIF2D (ligatin) in the reconstituted translation system (Skabkin et al., 2010; Skabkin et al., 2013; Zinoviev et al., 2015). Thus, DENR:MCT-1 and eIF2D (ligatin) may be more conducive to the lateral attachment of mRNA, prompting their further investigation in shunting-like events.

Some data suggested that shunting may be promoted by complementarity interactions between mRNA and rRNA (Matsuda & Mauro, 2014; Yueh & Schneider, 2000), which is another long-standing alternative paradigm of eukaryotic initiation (Chappell, Edelman, & Mauro, 2006; Dresios, Chappell, Zhou, & Mauro, 2006; Hu, Tranque, Edelman, & Mauro, 1999; Matsuda & Mauro, 2014; Mauro & Matsuda, 2016; Tranque, Hu, Edelman, & Mauro, 1998) that, however, has had limited experimental support. Recent structural studies directly reveal that limited base-pairing between SSU RNA (helix 16 of 18S rRNA near the entry to the mRNA-binding cleft) and nonpaired bases of the "three-way junction" structure of histone h4 mRNA (located downstream from the start codon) can occur (Martin et al., 2016). Histone h4 mRNA:18S rRNA interactions were proposed to provide direct tethering of the SSUs to the start site (Martin et al., 2016), thus contributing to the observed cap-dependent, but scanning-deprived, initiation on the h4 mRNA (Martin et al., 2011; Smith et al., 2005). The likely competition of the complementarity interactions with the DHX29 helicase binding site on the SSU (des Georges et al.,

2015; Pisareva et al., 2008) may explain why certain RNA structures can be more efficiently bypassed during scanning *in vivo* (Martin et al., 2016). Recent biochemical evidence prompts a resurfacing of the concepts of rRNA:mRNA base-pairing during scanning-based initiation, also at the mRNA exit side. Complementarity interactions between the termination upstream ribosome binding site (TURBS) and helix 26 of SSU rRNA (near the mRNA exit side) were shown to be required for the reinitiation on *Caliciviride* mRNAs (Luttermann & Meyers, 2009; Meyers, 2007). Notably, TURBS-mediated reinitiation was possible without eIF3 and was compatible with either canonical eIFs 1, 1A, 2, or eIF2D (ligatin) (Zinoviev et al., 2015).

Examples of shunting and possible lateral rebinding of SSUs have inspired an alternative to the scanning model. In the “mRNA looping” hypothesis, which can be viewed as an extreme case of nonlinear motion or shunting, SSU hopping from the 5′ cap to random locations in the 5′ UTR was suggested as a mechanism of systematic start codon search (Jang & Paek, 2016; Paek et al., 2015). While *in vivo* snapshots of scanning SSU distributions obtained with TCP-seq do not provide proof of elevated attachment to cap structures at the steady state, or prevalence of nonhomogeneous distribution of scanning SSUs (Figure 7a), cases of SSU pile ups were observed in multiple 5′ UTRs that were not previously attributed to initiation through uORFs (Archer et al., 2016). Thus, it will be critical to distinguish between potential internal ribosome entry, shunting, or pausing due to complementarity interactions with scanning SSUs in the future and to elucidate the extent of these events of initiation across transcriptomes in different *in vivo* cellular environments.

5 | RECOGNITION OF START CODONS AND JOINING BY THE LARGE RIBOSOMAL SUBUNIT

The third stage of cap-dependent initiation involves the recognition of the start codon of mRNA, arrest of SSU motion, and the subsequent joining of the LSU to assemble the elongation-capable ribosome, with its P-site occupied by Met-tRNA_i^{Met}. The latter serves as nascent “peptide” (in this case—just the methionine residue) donor in the first transpeptidation reaction. Thus, the whole first half of the start codon recognition process in eukaryotes substantially differs from the prokaryotic initiation, where SSUs do not have to transition from the powered 5′ → 3′ motion into a stall.

The consensus initiation model proposes pausing of the scanning SSUs over start codons located in a strong nucleotide context as the first step of start codon recognition. Pausing allows cognate codon–anticodon interactions to affect eIF2 and the small eIFs 1 and 1A, which trigger the transition of the SSU from its scanning-competent mode to its subunit-joining mode and thus control the fidelity of start codon recognition. The transition process begins with the hydrolysis of eIF2-bound GTP, dissociation of eIF1 from the complexes, and P_i release from eIF2:GDP (Figure 2). This is followed by a major conformational rearrangement of the SSU resulting in a more tight (“locked”) binding to mRNA, preventing further SSU sliding. The eIF2:GDP then leaves the complex, allowing eIF5B:GTP to bind to Met-tRNA_i^{Met} and SSU and begin the evolutionarily conserved subunit joining process. Because many of the start codon recognition complexes can be artificially stabilized and purified or assembled *in vitro*, there is now an extensive structural insight into the major start codon recognition stages (Figure 9). This is complemented by biochemical and kinetic data for the well-defined “checkpoints”, as well as many ribosome profiling datasets where initiation site frequencies were determined. Furthermore, *in vivo* SSU footprinting by TCP-seq demonstrates at least three different complex conformations over start codons, matching some of the *in vitro* conclusions (Figure 9a) (Archer et al., 2016). The most mechanistically intriguing part of the process remains the initial pausing and the transition of the moving SSUs to a complex that is almost irreversibly fixed at start codons. Otherwise, the start codon recognition appears as one of the better-understood initiation stages mechanistically, with recent structural works filling in the fine details of its substeps.

5.1 | Rearrangement of the scanning complexes

Compared to “scanning” SSUs, the structures of the early-stage start codon recognition complex prominently feature rearrangements of the position of Met-tRNA_i^{Met} in the P-site. Already in the beginning of start codon recognition, the entire TC appears to relocate closer to the mRNA in the SSU, resulting in a more stable binding and more tight interactions between SSU, Met-tRNA_i^{Met}, and mRNA. In several cryo-EM reconstructions of complexes stabilized with nonhydrolysable GTP analogues, SSUs transition from the P_{OUT} state characteristic to mRNA binding and/or scanning to the antagonized “P_{IN}” state. In the P_{IN} state, Met-tRNA_i^{Met} is positioned ~7 Å deeper in the P-site, with the 3′ acceptor stem relocated upward and toward the A-site. This relocation is stabilized by Met-tRNA_i^{Met} interactions with eIF2γ, as well as by the displacement of tRNA from E-site area by eIF2α (Figure 9; compare b1 and c1 with b2 and c2) (Hussain et al., 2014; Llacer et al., 2015). This Met-tRNA_i^{Met} conformation begins to resemble some prokaryotic initiation complexes, as was noted for the archaeal SSU complex (Hussain et al., 2014; Schmitt et al., 2012; Simonetti et al., 2016). In the P_{IN} state, the three conserved G:C

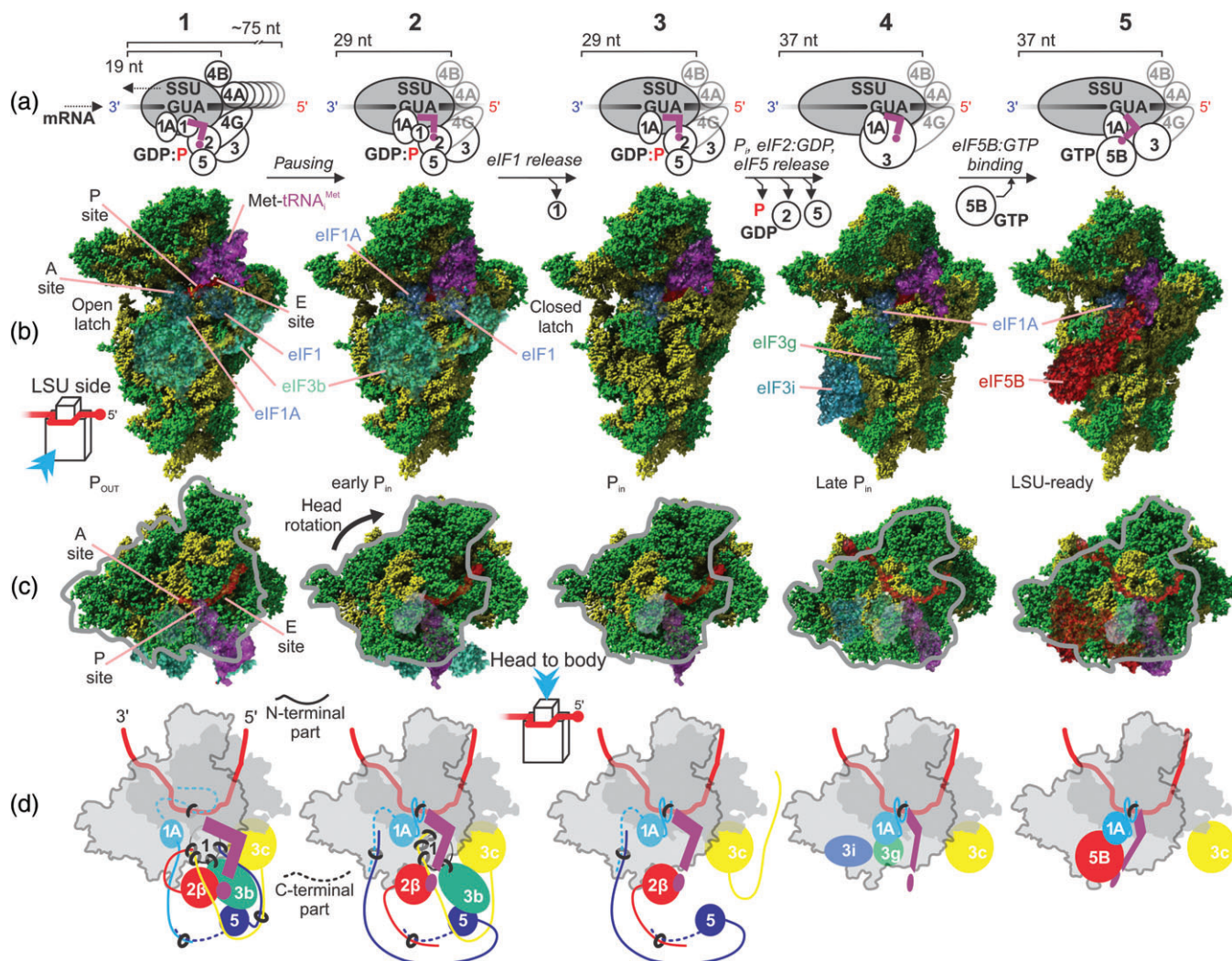


FIGURE 9 Possible configurations of the initiating SSUs as they progress through the start codon recognition stages (1–5). (a) Schematics depicting the possible changes in composition of initiation complexes and matching *in vivo* TCP-seq footprint lengths (Archer et al., 2016; Shirokikh, Archer, Beilharz, Powell, & Preiss, 2017) (see text). (a1) represents initial pausing over the start codon, while the SSU remains in its scanning configuration featuring the open latch, the head swiveled to the solvent side, and the Met-tRNA_i^{Met} displaced laterally from the P-site in the scanning-compatible P_{OUT} conformation; (a2) depicts an early stage of start codon recognition where the mRNA latch is tightened and the mRNA entry more constricted, with Met-tRNA_i^{Met} moving further into the P-site and partially acquiring P_{IN} state, which induces delocalization of eIF1 within the complex; (a3) shows the stabilized P_{IN} with the Met-tRNA_i^{Met} strongly bound in the P-site, resulting in the full arrest of SSU sliding on mRNA as eIF1 leaves the initiation complex; (a4) depicts the initiation complex after phosphate release gated by the eIF1 detachment, also resulting in the release of eIF2:GDP and eIF5 from the SSU; (a5) shows the last stage of the process where the Met-tRNA_i^{Met} is rebound by eIF5B:GTP, resulting in a readjustment of its position, making it compatible with productive LSU attachment. (b) Structures of the SSU reorganization viewed from the LSU side and corresponding to the start codon recognition stages as shown in (a), with the focus on the intersubunit surface rearrangements. Densities of only mRNA, Met-tRNA_i^{Met}, eIFs 1, 1a, 3b, 3i, 3 g, and 5B are shown for simplicity. SSU depicted as described in Figure 8a. (b1) Structure of the yeast initiating SSU in the open state (PDB entry 3JAP) (Llacer et al., 2015) with eIF3b (PDB entry 5K1H) (Simonetti et al., 2016) modeled as described in Figure 8a. mRNA model is aligned from the closed state (PDB entry 3JAP) by the 18S rRNA trace in the body region of the SSU. (b2) Same as (b1) but for the SSU in the closed state (PDB entry 3JAP) (Llacer et al., 2015). (b3) Same as (b2) but with the densities of eIF1 and eIF3b omitted. (b4) Structure of the rabbit late P_{IN} SSU (PDB entry 5K0Y) (Simonetti et al., 2016) with the eIF1A density modeled from the yeast closed state (PDB entry 3JAP), by aligning the 18S rRNA trace in the body region of the SSU. (b5) Structure of the rabbit SSU complexed with eIF5B:GMP-PNP in the PRE-translocation-like state (PDB entry 4UJD) (Yamamoto et al., 2014) during the late start codon recognition stage (derived from the complete ribosome structure by removing LSU-related densities). mRNA and eIF1A models aligned from the yeast closed state SSU (PDB entry 3JAP) by the 18S rRNA trace in the body region. (c) Same as (b) but in the head-to-body view with the head density shown as semitransparent and with gray outline. (d) Schematics of the key interactions between the LSU-side factors progressing through the start codon recognition stages corresponding to the models in (a)–(c). Head-to-body view as in (c) schematized as in Figure 3. (d1) Multiple interactions between LSU-side eIFs guided by the C- and N-terminal parts of eIF1A and eIF5 converge on eIF1 binding and stabilize the scanning configuration. (d2) The eIF1A C-terminal part is displaced from the P-site upon early P_{IN} state, leading to the structuring of its N-terminal part in the A- and P-sites and stabilization of the codon–anticodon interactions. Interaction of the liberated eIF1A C-terminal part with the N-terminal part of eIF5 and the liberated eIF5 C-terminal part with the eIF2β destabilize eIF1 binding. (d3–d5) The eIF1A- and eIF5-guided cascade of rearrangements leads to the loss of affinity to eIF1 and adoption of the stable scanning-arrested P_{IN} configuration and gated release of P_i, followed by the eIF2:GDP and eIF5 dissociation

base pairs of the Met-tRNA_i^{Met} anticodon stem make specific contacts with rRNA bases, which is suggested to be one of the factors responsible for the selective Met-tRNA_i^{Met} binding (Hussain et al., 2014; Lomakin & Steitz, 2013).

The mRNA path during start codon recognition appears similar to the “scanning” complexes; the bases of nucleotides adjacent to the start codon are unstacked from their neighbors and can make contacts with the P-site rRNA (Hussain et al., 2014; Lomakin & Steitz, 2013). The mRNA “latch” formed by the rRNA helices 18 and 34 at the entry side of the mRNA-binding cleft is closed in all structures modeling early start codon recognition. The overall mRNA-binding cleft configuration appears tighter, and the P-site arrangement is more compact in the P_{IN} state (Llacer et al., 2015; Simonetti et al., 2016). The head of the SSU is rotated clockwise by a further ~5°, which is proposed to facilitate a more stable TC binding (Hussain et al., 2014; Llacer et al., 2015; Lomakin & Steitz, 2013). The SSU head was reported to be tilted (swiveled) toward the SSU body at the intersubunit side for ~4°, further narrowing the mRNA-binding cleft in the late start codon recognition complexes (Simonetti et al., 2016).

When eIF1 is present in the complexes, it is slightly displaced away from Met-tRNA_i^{Met} and may induce electrostatic repulsion between its conserved negative residues and tRNA, but it still maintains contact with the P-site base pairs (Hussain et al., 2014; Llacer et al., 2015). Notably, eIF1A makes stacking interactions with the +4 mRNA nucleotide (adjacent to the AUG from 3'-side in the A-site), which is an important start codon context determinant. The eIF1A N-terminal part, which is unstructured in the scanning complexes, becomes stabilized and approaches the codon–anticodon pair from the A-site, suggestive of its role in P_{IN} stabilization (Hussain et al., 2014). The eIF1A C-terminal extension, in contrast, becomes delocalized and possibly uncoupled from eIF1 and Met-tRNA_i^{Met} (Hussain et al., 2014). Interestingly, in the start codon–SSU complexes purified from mammalian cell lysate and also stabilized with nonhydrolysable GTP analogue, densities could not be attributed to eIF1 and eIF1A (Simonetti et al., 2016). Thus, *in vitro* reconstruction with an excess of eIF1 and eIF1A may be relevant to an earlier and relatively less stable intermediate immediately prior to the dissociation of these factors. eIF2 primarily interacts with the SSU head (Llacer et al., 2015; Simonetti et al., 2016) in the start codon recognition complexes. eIF2α as part of the TC is located deep in the E-site and can make direct contacts with mRNA residues –3 and –2 (Hussain et al., 2014; Llacer et al., 2015; Simonetti et al., 2016). Notably, unlike in the scanning complexes, eIF2β repositions toward, and makes direct contacts with, the SSU head, additionally interlocking mRNA in the SSU mRNA-binding cleft. At the same time, eIF2β loses its eIF1 and eIF1A interactions, suggestive of its role in the stabilization of the scanning configuration where it interacts with these factors (Llacer et al., 2015).

A lateral displacement of the core PCI/MPN eIF3 subunits was reported to coincide with the P_{IN} state, suggesting large-scale molecular motions of eIF3 during start codon recognition (Llacer et al., 2015; Simonetti et al., 2016). Although there is some discrepancy in the attribution of the electron density, it was suggested that eIF3i and eIF3g are present at the entry side below the mRNA-binding cleft on the intersubunit surface of the SSU and may contact eIF2γ upon start codon recognition. Thus, eIF3i and eIF3g may actively participate in the stalling of scanning SSUs at start codons (Llacer et al., 2015). Alternatively, eIF3b also relocates to the intersubunit SSU side below the A-site from the solvent-side entry and makes contacts with eIF2γ until the early start codon recognition stage (Figure 9 b1,2, c1,2). This eIF3b relocation is hypothesized to additionally stabilize eIF2β interactions with eIFs 1 and 1A and be conducive to scanning (Simonetti et al., 2016). Upon establishing the complete codon–anticodon interactions and the likely release of eIF1, eIF3b binding may become destabilized, and approximately the same position could be retaken by the eIF3i and eIF3g subunits (Figure 9 b4, c4) (Simonetti et al., 2016). Therefore, replacement of eIF3b with eIF3i and eIF3g may be one of the major structural events during start codon recognition and signify the establishment of stable codon–anticodon interactions.

Notably, in the mRNA-free SSUs, neither eIF3b nor eIF3i and eIF3g were observed in this location and, instead, were localized near the mRNA entry at the solvent side, and eIF3j was reported to bind near the intersubunit entry side (Figure 5d) (Aylett et al., 2015; des Georges et al., 2015; Erzberger et al., 2014). Thus, eIF3 may remain attached to the SSU through eIF3i and eIF3g at least until the subunit joining step, with a possible role of these eIF3 subunits in the interim Met-tRNA_i^{Met} stabilization and eIF5B binding control (Simonetti et al., 2016). There is limited structural evidence that eIF5 may be present in the early P_{IN} complexes closer to the A-site and may maintain contacts with eIF1 and eIF2γ (Figure 4g) (Hussain et al., 2014).

The observed structural rearrangements immediately upon start codon recognition correspond well with the biochemical evidence. Dissociation of eIF1 was shown as one of the key steps during start codon recognition (Cheung et al., 2007). The role of eIF1A C- and N-terminal of eIF1A extensions, and their sequential relocation, was extensively studied by mutagenesis and localization approaches (Fekete et al., 2007; Martin-Marcos, Cheung, & Hinnebusch, 2011; Mitchell & Lorsch, 2008; Saini et al., 2010). Involvement of eIF2α in the stabilization of the –3 nucleotide of mRNA (adjacent to the start codon from E-site) was shown (Pisarev et al., 2006; Visweswaraiiah & Hinnebusch, 2017), as well as eIF2β contacts with eIF1 and Met-tRNA_i^{Met} (Algire, Maag, & Lorsch, 2005; Martin-Marcos, Cheung, & Hinnebusch, 2011). The binding of eIF3 to the SSU through its “peripheral” subunits was previously demonstrated (Fraser et al., 2004; Phan, Schoenfeld,

Valasek, Nielsen, & Hinnebusch, 2001; L. S. Valasek, 2012; Zhou et al., 2008), as well as its involvement in the fidelity of start codon recognition (Asano, Phan, Anderson, & Hinnebusch, 1998; Karaskova et al., 2012; Phan et al., 2001). The eIF5 N-terminal part was shown to bind eIF2 and structurally mimic eIF1, antagonizing its binding. Its C-terminal part was observed to exclusively interact with eIF1 or the N-terminal part of eIF2 β and participate in the control of eIF2 release (Lorsch & Dever, 2010; Majumdar & Maitra, 2005; Saini et al., 2015; L. S. Valasek, 2012). The C-terminal part of eIF5 was also shown to interact with eIF3c (Karaskova et al., 2012). More recent evidence includes the involvement of eIF3c interactions with limited eIF1 surfaces and the C-terminal part of eIF5 during scanning, which cease on start codon recognition and result in an extended eIF3c attachment to eIF1 incompatible with eIF1 SSU binding (Obayashi et al., 2017).

The exit part of the mRNA-binding cleft, together with eIF2 α , was demonstrated to contribute to the P_{IN} stabilization at AUG codons (Visweswarajah & Hinnebusch, 2017), and a lesser accessibility of the entry side of the mRNA-binding cleft and P-site was shown (Zhang et al., 2015). Thus, an overall sequence of events can be reconstructed, where the scanning SSUs first temporarily stall (pause) over the start codons, assisted by strong codon–anticodon interactions and additionally by binding of, at least, -3 and $+4$ mRNA nucleotides to rRNA and eIF2 α or eIF1, respectively (Figure 9 a1–d1). These interactions restructure the open scanning SSU configuration to the early closed P_{IN} state. The eIF1A N-terminal part becoming structured in the A-site and near the codon–anticodon interactions in the P-site appears as the main driver of this rearrangement, yielding a more stable complex (Figure 9 a1–d1). The eIF1A N-terminal part also loses its affinity to the C-terminal part of eIF5, which induces eIF5 binding to the N-terminal part of eIF2 β . The eIF5 C-terminal:eIF2 β N-terminal interaction displaces eIF1 from its binding site on eIF5. The eIF1A C-terminal part becomes displaced from the P-site and may interact with the N-terminal part of eIF5 (Figure 9 a2–d2). This results in eIF1 displacement, which may be further promoted by eIF3c binding to the eIF1 surfaces that are required for its interaction with the SSU. eIF1 displacement concludes the early start codon recognition stage. The subsequent eIF1 dissociation from the complex triggers P_i release from eIF2:GDP: P_i , and the TC, together with eIF5, loses affinity to the SSUs (Figure 9 a3–d3). Dissociation of eIF2 enables displacement of eIF3b from the intersubunit surface, and eIF3i and eIF3g may rebind in the vicinity of Met-tRNA Met , preventing premature LSU attachment or rebinding of eIF2:GTP in the “late” P_{IN} SSU. eIF3i and eIF3g possibly additionally stabilize the “post-eIF5” configuration of the SSU (Figure 9 a4–d5).

One of the remaining mysteries of start codon recognition is the exact timing and trigger of eIF2-bound GTP hydrolysis and the mechanism of P_i release. In the absence of the corresponding structures, it is not fully clear whether eIF5 acts as a classical GTPase by providing an “arginine finger” or whether it stimulates eIF2’s own GTPase activity (Cheung et al., 2007; Jackson et al., 2010; Marintchev & Wagner, 2004). More importantly, it was suggested that GTP hydrolysis may occur already *en route* during scanning, with start codon recognition triggering eIF1 and, consequently, P_i and eIF2:GDP release (Algire et al., 2005). GTP hydrolysis step is an “irreversible” process, and thus, it could be preferred as the driver for the large-scale rearrangements, as in many known G-proteins. However, the strict requirement of the initial rapid response at start sites could have modified this evolutionary conserved mechanism toward a somewhat more reversible but faster-acting pathway. Hypothetically, the GTP hydrolysis rate of $\sim 5\text{ s}^{-1}$ observed in the scanning complexes could contribute to the known skipping of 5′-proximal start codons, owing to the comparably fast ($\sim 10\text{ nt/s}$) scanning motion (Algire et al., 2005; Berthelot et al., 2004; Kozak, 1991; Pestova & Kolupaeva, 2002; Vassilenko et al., 2011). In this regard, it is also noteworthy that up until the eIF5B:GTP (and possibly LSU) attachment, start codon recognition is not irreversible in a strict sense. For example, it was demonstrated that eIF1 can destabilize weaker P-site mRNA:Met-tRNA Met interaction after GTP hydrolysis and the departure of eIF2 (Lomakin et al., 2006). Similarly, it was shown that SSUs in the P_{IN} state can be spontaneously displaced from the start codons and resume scanning toward the more downstream triplets (Terenin et al., 2016). These results suggest that at least up until eIF5B attachment, SSUs do not fully lose the ability to scan mRNA and that prolonged pausing at start sites, if not concluded with eIF2 dissociation, may deteriorate the fidelity of start codon recognition, rather than improve it.

Intiguously, TCP-seq demonstrates three predominant SSU footprints at start codons where the major protection change occurs over the inbound (3′, entry) portion of the mRNA (Figure 7) (Archer et al., 2016). Using the extended 5′ protection of footprints characteristic for scanning SSUs (see above in the Scanning section) and an accumulation of queued SSUs in front of the start codons, it was possible to assign these footprints to early, middle, and late stages of start codon recognition. Importantly, in full concordance with the other evidence, these complexes showed increasing length of protection of the entry mRNA portion over time. Thus, it is possible that several key start codon intermediates with relatively long half-lives (and thus important for the overall initiation rates) can be monitored and quantitatively assayed for each mRNA *in vivo*. The sub-second dynamics of the eIF1 displacement (not dissociation) makes these steps likely transparent to TCP-seq (Maag et al., 2005), and the presence of the 5′-extended protection allow the suggestion that the first *in vivo* intermediate is a paused scanning SSU still in the P_{OUT} configuration (protecting overall just $\sim 19\text{ nt}$ of mRNA; Figure 9 a1). The relatively slow eIF1 dissociation and P_i release ($\sim 2\text{ s}$) (Maag et al., 2005) suggest that the second stage represents complexes in the early P_{IN} state

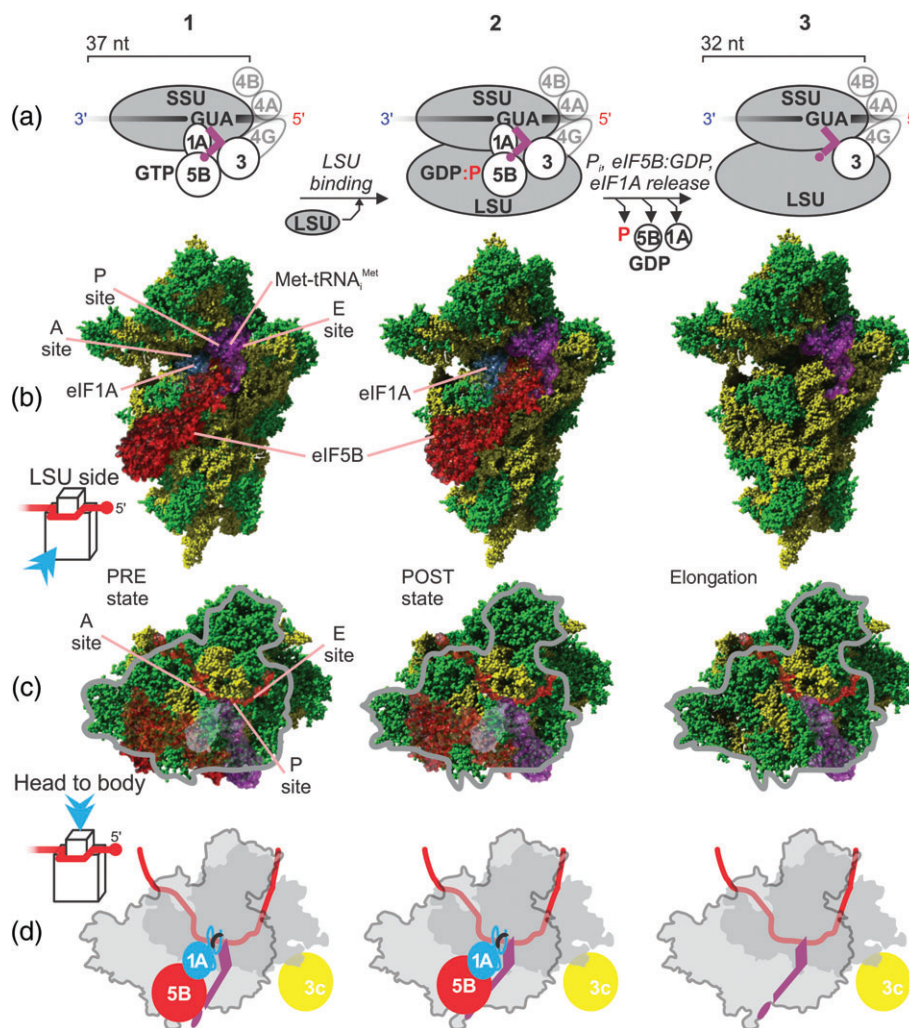


FIGURE 10 Possible configurations of the initiating SSU as it progresses through the postulated stages of LSU joining. Figure layout as described for Figure 9. (a1) represents PRE-translocation-like state of the SSU:(GTP:eIF5B:Met-tRNA_i^{Met}) complex when eIF5B:GTP first attaches to the late-P_{IN} SSUs and reorients the Met-tRNA_i^{Met}. This complex may now productively bind LSU. (a2) depicts POST-translocation-like state of the SSU:(GTP:eIF5B:Met-tRNA_i^{Met}) complex, adopted upon attachment to the LSU and ~6° SSU back-rolling on it along its head-to-body vector, fully slotting the Met-tRNA_i^{Met} into the LSU P-site. This stage may coincide with the hydrolysis of the eIF5B-bound GTP. (a3) represents the first elongation intermediate, with the complete ribosome protecting ~32 nt over the start codons as determined by the *in vivo* TCP-seq footprinting (Archer et al., 2016; Shirokikh et al., 2017). (b1) structure of the rabbit SSU complexed with eIF5B:GMP-PNP in the PRE-translocation-like state (as in Figure 9b5). (b2) structure of the rabbit SSU complexed with eIF5B:GMP-PNP in the POST-translocation-like state (PDB entry 4UJC) (Yamamoto et al., 2014). mRNA and eIF1A modeled as described for Figure 9b5. (b3) hypothetical rabbit SSU structure corresponding to the first elongation intermediate, derived from (b2) by removing the eIF5B- and eIF1A-corresponding densities. (c) Corresponds to (b) but from the head-to-body point of view. (d) Cartoon schematics corresponding to (c) depicts as described in Figure 3

up until the dissociation of eIF2. These early *in vivo* “P_{IN}” complexes protect mRNA by a further 10 nt in 3′ direction (Figure 9 a2, a3). The third type of *in vivo* start codon complexes has the most extended 3′ protection, longer by 8 nt than the middle type and protecting ~37 nt overall (Figure 9 a4, a5). This 3′-ward increase in protection may represent rearrangements of the peripheral eIF3 subunits and the overall more “prokaryotic” position of Met-tRNA_i^{Met} as it leans toward A-site in the late P_{IN} complexes. Alternatively, the longest SSU footprints over the start codons could be explained by eIF5B binding (see below), given that *in vitro*-measured dynamics of eIF5B association and dissociation are relatively slow (Acker et al., 2009), or be the result of a mixture of both of these processes. Importantly, from these data, the start codon recognition stage also clearly appears as rate-limiting for ~65% of the quantified mRNAs, based on the ratio of the start codon SSU pileups versus scanning SSU pileups in the 5′ UTR. The data also suggest that the dynamics of different start codon recognition substeps may vary substantially across the transcripts. Thus, start codon recognition may play a somewhat unexpected role where it additionally shapes the speed of translation initiation in an mRNA-selective way. The exact determinants of this, and the actual eIF composition of the *in vivo* start codon intermediates, remain to be elucidated.

5.2 | Assembly of elongation-capable ribosomes

Initiation concludes with the eIF5B-assisted joining of the LSU (Figure 10a). Model structures of ribosomes at the start codon where Met-tRNA_i^{Met} is bound to eIF5B stabilized with nonhydrolysable GTP analogues were obtained for yeast (assembled with “nonstructured” [U,C]₂-containing synthetic mRNA, SSUs, LSUs, Met-tRNA_i^{Met}, eIFs 1, 1A, 2:GTP, 5 and 5B:GMP-PCP) (Fernandez et al., 2013) and mammalian (mRNA containing HCV IRES, SSUs, LSUs, Met-tRNA_i^{Met}, eIFs 3 and 5B:GMP-PNP) (Yamamoto et al., 2014) initiation systems and demonstrate highly similar overall arrangements, also reminiscent of bacterial initiation complexes (e.g., Hussain, Llacer, Wimberly, Kieft, & Ramakrishnan, 2016). Notably, a structure of eIF5B bound to the SSUs alone have not yet been obtained (although bacterial complexes with analogous configuration exist) but may represent another intermediate of initiation. Another point of view is that there is no such distinct intermediate in eukaryotes, and all three components (SSUs with mRNA, Met-tRNA_i^{Met}, and the residing eIFs, LSUs, and eIF5B:GTP) can only form unstable bilateral complexes with relatively short half-lives, which only become productive upon the binding of the third partner. On the SSU side of the ribosome:eIF5B complexes (Figure 10b–d), eIF5B adopts conformation markedly distinct from its “free” form, with its tRNA-binding domains more extended and reaching anticodon stem loop from the main ribosome-binding side located peripherally on the intersubunit surface down from the mRNA-binding cleft entry toward the middle of the SSU body (Figure 10b,c). eIF5B makes multiple contacts with the SSU, as well as with the LSU, as could be expected from its subunit “bridging” function (Fernandez et al., 2013; Yamamoto et al., 2014). The overall conformation of the SSU itself remains unaltered substantially compared to the previous steps, but the positions of Met-tRNA_i^{Met} and SSU relative to the LSU are considerably different (Fernandez et al., 2013; Yamamoto et al., 2014). In the yeast complexes, the SSU is rotated slightly (~3.5°) counterclockwise relative to the LSU along the axis perpendicular to the intersubunit plane. Met-tRNA_i^{Met} is stabilized in a position different from the regular P-site-bound tRNA and appears in a “hybrid” (“P/I”) state, overall leaning toward the A-site with the CCA 3′ end bent toward the mRNA entry side by eIF5B, and the tRNA body is located generally closer to the SSU than in the prokaryotic complexes (Figures 9–10) (Fernandez et al., 2013; Yamamoto et al., 2014). Interestingly, the reconstruction in the mammalian system resulted in subpopulations of complexes demonstrating SSU “rolling” (rotation along the axis aligned with the head-to-body direction and the intersubunit plane) relative to the LSU, which is unique to eukaryotes (Yamamoto et al., 2014). Relating these complexes to the pre- and post-translocation elongating ribosomes allowed the inference of the likely mechanism of the eukaryotic subunit joining. The initial binding occurs with Met-tRNA_i^{Met} not fully accommodated in the LSU P-site (with Met-tRNA_i^{Met} CCA 3′ located in the LSU A-site) but allowing its slotting into the LSU cavity without affecting the complex geometry, followed by the entire SSU assembly rolling “back” (~6° clockwise around the head-to-body vector) on the LSU, which results in the full accommodation of the Met-tRNA_i^{Met} in the ribosomal P-site in a conformation compatible with the subsequent elongation phase. The “reverse rolling” likely immediately precludes or follows eIF5B-bound GTP hydrolysis and was also suggested as the main driver of the eIF5B affinity loss and its release from the ribosome (Yamamoto et al., 2014). Thus, while eukaryotic ribosomes seem to utilize a unique mechanism of the subunit joining, eIF5B function in adjusting Met-tRNA_i^{Met} conformation remains well conserved across all life.

Whereas the mechanics of subunit joining seem to be well understood, certainty is lacking for the time points of the release of eIFs that remain associated with the ribosome at these stages. After the initial start codon recognition stages (Figure 9), the SSUs are presumably left with eIF3 (contacting the intersubunit area below A-site with its eIF3i and eIF3g subunits and PCI/MPN core) and, possibly, the eIF4-group factors and eIF1A. This transition state is one of the remaining structurally uncharacterized intermediates, and it is unclear how Met-tRNA_i^{Met} position is stabilized in these complexes. The fact that such a complex cannot be efficiently obtained *in vitro* by mixing the respective factors and an unstructured mRNA suggests that either mRNA remains tightly interlocked and thus cannot get out (and in) of the SSU or that these complexes are highly unstable. *In vitro* purification of the post-eIF5 complexes supports the first alternative (Unbehaun et al., 2004). As it was suggested, eIF3 can provide a ribosome antiassociation activity at this stage (Simonetti et al., 2016). Density has not been attributed to eIF1A in these complexes, and thus, it may be concluded that it is unstably bound or dissociated at the stages of the eIF5B attachment. Similar to this, and to the situation with the sequence of eIFs attachment during cap recognition, there is no certainty whether eIF5B first attaches to the SSU in the closed state followed by the LSU attachment or whether the LSU first binds to eIF5B prior to subunit joining, although mechanistically, this is a lower principal question to the regulation of the initiation outcome and rates. *In vitro* experiments suggest that the presence of LSU is not mandatory for the eIF5B attachment; LSUs do not efficiently attach without eIF5B (Pestova et al., 2000; Pestova et al., 2008). In this regard, it is peculiar that eIF5B is a nonessential gene in yeast (Choi, Lee, Zoll, Merrick, & Dever, 1998). Thus, either subunit joining has more relaxed eIF requirements in lower eukaryotes, or other mechanisms or factors may be involved. It is also incompletely understood when eIF3 and eIF4 group factors leave the ribosomes. Indeed, whereas eIF3 “peripheral” subunits may completely detach from the intersubunit SSU side, detachment of its “core”, although it may become more laterally oriented, is not certain. In this low-affinity state, eIF3 may remain attached to the elongating ribosomes for some time, which

was originally suggested as one of the mechanisms explaining reinitiation after short ORFs (Cuchalova et al., 2010; Mohammad, Munzarova Pondelickova, Zeman, Gunisova, & Valasek, 2017; Poyry et al., 2007). In the case of mammalian ribosomes, this enables a possibility of eIF4 group factors remaining on the ribosomes as well and, in case of yeast translation, raises a question of what mechanism controls eIF4G release. As was discussed above, the most 3'-extended *in vitro* SSU footprints obtained with TCP-seq can also be a result of eIF5B binding (Archer et al., 2016). Thus, in the future, TCP-seq targeted at the SSUs and complete ribosomes containing eIF5B, eIF3, and group 4 eIFs could be used to provide a more definitive answer to the sequence of events and discern when eIF5B becomes associated with the initiation complexes and when eIFs 1A, 3, and 4-group leave the ribosomes *in vivo*.

5.3 | Determinants of start codon recognition

The prevalent start codon in eukaryotes is the methionine-encoding AUG, which is recognized by a special Met-tRNA_i^{Met} during initiation (Dong et al., 2014; Kolitz & Lorsch, 2010; Lomakin et al., 2006). Nevertheless, AUG is used as the start for only ~50% of coding regions in mammals (Ingolia, Lareau, & Weissman, 2011), as compared to ~82% in bacteria (Hecht et al., 2017). While these proportions may be somewhat affected by nonequilibrium effects of the ribosome profiling technique, it is evident that eukaryotic start site selection is far more complex than initially thought. The use of alternative start codons can be classified either by the so-called “near-cognate” interactions within otherwise identical, canonical initiation through scanning or by alternative initiation pathways employing special components and a different tRNA, such as it was exemplified by eIF2A and Leu-tRNA_c^{Leu} (Starck et al., 2016), where perfect base-pairing is achieved at the non-AUG (CUG) codon. Near-cognate start codons may often involve imperfect base-pairing where the first nucleotide of the codon noncanonically interacts with the third U residue of the Met-tRNA_i^{Met}, such as in the case of UUG, GUG, and CUG triplets (Ingolia et al., 2011). The same triplets appear as the most preferred near-cognate codons in bacteria (Hecht et al., 2017). In eukaryotes, however, other near-cognate start codons with substitutions in the second and third nucleotides are encountered relatively frequently and cumulatively contribute to the less frequent use of AUG (Ingolia et al., 2011; Lind, Esguerra, & Aqvist, 2017). In this regard, a remarkable difference is that the GUG and UUG near-cognate codons dominant in prokaryotes are much less utilized in eukaryotic initiation, with the evidence of selective GUG and AAG suppression by eIF1 and AUA suppression by both eIF1 and eIF1A. In contrast, CUG is not suppressed and, with an occurrence rate of ~20%, can be considered a true eukaryotic start site alternative (Ingolia et al., 2011; Lind et al., 2017; Lind & Aqvist, 2016). Given that eukaryotic SSUs scan long sequences of nucleotides, such selective suppression of the stronger-binding triplets appears to be an evolutionary mechanism responsible for the highly precise eukaryotic start site selection (Asano, 2014; Lind & Aqvist, 2016; Lind et al., 2017).

Critically, the transition from the scanning-competent mode into the arrested mode at the start codon must be a relatively fast process to be compatible with scanning speeds of ~10 nt/s (Karine Berthelot et al., 2004; Vassilenko et al., 2011). The fidelity of codon recognition thus becomes paramount, and the more complex eukaryotic machinery displays features developed specifically to facilitate “on-the-fly” recognition of cognate *versus* false start sites (Asano, 2014; Lind & Aqvist, 2016; Lomakin et al., 2006; Lorsch & Dever, 2010). The readiness of the scanning complex itself during initiation on various mRNAs is one of the key determinants of start codon recognition capability, which can be programmed by the levels of various active eIFs accessible to the initiating SSUs at that moment. For instance, increased eIF1 levels or decreased TC levels can promote more “scan-through” events, and weak start sites will be skipped more frequently (Fijalkowska et al., 2017; Hinnebusch et al., 2004; Ivanov, Loughran, Sachs, & Atkins, 2010). The controlled (increased or decreased) “leaky scanning” through start sites, combined with uORFs, upstream start codons, and alternative start sites, provides a vast regulatory potential (Fijalkowska et al., 2017). However, for this system to work and from a gene-wise perspective, the key start site determinants are those encoded by the sequence (and structure) of the mRNA. Classically, the “consensus” nucleotide “context” or “Kozak consensus” sequence is considered to be the second important start site determinant, after the start codon itself. For mammals, this consensus sequence is usually referred to as (G₋₉C₋₈C₋₇G₋₆C₋₅C₋₄)Pu₋₃C₋₂C₋₁A₊₁U₊₂G₊₃G₊₄ (Kozak, 1987) and differs substantially across taxa. For example, the yeast consensus is considered to be (A/U₋₆A₋₅A/C₋₄)A₋₃A₋₂A/C₋₁A₊₁U₊₂G₊₃ (U₊₄C₊₅Py₊₆) (Miyasaka, 1999), and the consensus in higher plants is A₋₃(A/C)₋₂A₋₁A₊₁U₊₂G₊₃G₊₄C₊₅ (Joshi, Zhou, Huang, & Chiang, 1997). These sequences were derived from a compilation of relatively abundant, well-translated mRNAs. Current genome-wide data suggest less-stringent context criteria and more diverse combinatorial possibilities, the rules of which remain to be outlined in detail (Fijalkowska et al., 2017; Hellens, Brown, Chisnall, Waterhouse, & Macknight, 2016; Ingolia et al., 2009; Ingolia et al., 2011; Michel, Andreev, & Baranov, 2014).

While the mechanistic aspects of the SSU reconfiguration at start codons are well understood, perhaps one of the largest remaining mysteries in start codon recognition is the exact mechanism by which the start codon-adjacent nucleotide context is influencing different stages of the process. Conceptually, several possibilities can be outlined. First, by altering the affinity of the initiating SSU to mRNA, the start codon context can affect the speed of scanning. For example, slower scanning due

to the downstream structure has been originally suggested to explain more efficient start codon recognition (Kozak, 1990). Interactions of the N-terminal part of eIF1A and the helix 44 of SSU rRNA with the nucleotide next to the P-site (position +4), were proposed to contribute to mRNA pausing during scanning (Lomakin & Steitz, 2013). Both +4 and −1 nucleotides are unstacked from the adjacent mRNA nucleotides and are interacting with rRNA (−1) and eIF1A (+4), and these, as well as nucleotide −3 interactions with different initiation components (*e.g.*, eIF1A, A/P/E-site rRNA, eIF2 α , *etc.*) (Hussain et al., 2014; Lomakin & Steitz, 2013), were further proposed as the factors contributing to the SSU pausing. In particular, evidence exists of eIF2 α interactions with nucleotides in positions −2, −3, which are thought to be critical for successful start codon context recognition (Pisarev et al., 2006; Visweswaraiah & Hinnebusch, 2017). Second, start codon context, by forming specific (but not necessarily strong) interactions with the components of the initiating SSU, can signal for the rearrangement of the complex, triggering faster transition to the next stage. For example, such specific signals may speed up the release of eIF1 and eIF2 and thus force the scanning SSU into the closed state faster. Less evidence has been accumulated for this possibility; however, this may be due to the transient nature of the involved intermediates, rather than their absence. A combination of the pausing and triggering mechanisms may be a third alternative.

It should be mentioned that, while generally widespread, start codon context may not be present on some mRNAs. The simplest examples of this are mRNAs with extremely short, or completely lacking, 5' UTRs (Akulich et al., 2016; Maicas, Shago, & Friesen, 1990). SSUs may still be binding to mRNA through regular eIF4E:cap interactions; however, no scanning commences during initiation, and thus, there is no requirement for nucleotide context to interact with the scanning SSUs. Another case of such a functional replacement of the start codon context is the Translation Initiator of Short 5' UTR (TISU) element C/G_{−4}A_{−3}A_{−2}G/C_{−1}A₊₁U₊₂G₊₃G₊₄C₊₅G₊₆G₊₇C₊₈, a start site configuration often found in well-translated mRNAs with extremely short 5' UTRs (as little as 4 nt 5' UTR length with median ~12 nt) (Elfakess & Dikstein, 2008; Haimov et al., 2015). Detailed structural explanations for TISU function are still to be discovered, but this is a physiologically relevant motif shown to be frequent in mitochondria- and metabolism-related mRNAs, translation of which is not repressed under energy-deficient conditions, in contrast to the majority of canonically scanned mRNAs (Sinvani et al., 2015). The TISU element is not scanned as it can be inferred from its unique stimulation by eIF1, which normally would increase the bypass of the start codons (Elfakess et al., 2011). Instead, it is hypothesized that the TISU element can stimulate eIF4F release from the initiation complex once the mRNA is bound to the SSU, eliminating scanning and possible eIF4F interference with a start codon located close to the mRNA 5' end (Sinvani et al., 2015).

In vivo TCP-seq SSU footprinting over the start codons provides an intriguing, and somewhat unexpected insight, in this regard. Even when SSU footprints over start codons are compared only between the highly efficiently initiated mRNAs (as can be measured by the high ratio of the start codon SSU pile-ups vs. pile-ups in the 5' UTRs) (Archer et al., 2016; Shirokikh et al., 2017), different mRNAs demonstrate distinctively different footprint patterns (Figure 7). Accounting for the attribution of the different start codon recognition stages to the different footprint 3'-end extensions over the start codons (as explained in Figure 9 and the Rearrangement of the Scanning Complexes section), it can be suggested that the proportions of the SSU dwell time in each state vary across mRNAs. Thus, start codon recognition may be fine-tuned more acutely than previously thought, and its control may involve not only pausing over the start sites but also signals in mRNA that affect the dynamics of subsequent stages, urging for further investigations of the underlying mechanisms.

6 | CONCLUSION

Recent times have seen outstanding progress in deciphering the molecular mechanism of eukaryotic translation initiation by cap-dependent ribosome recruitment and 5' UTR scanning. Many of the original findings obtained through hundreds of painstaking biochemical and genetic studies have been substantiated by insight derived from high-resolution crystallographic and cryo-EM structures of the small ribosomal subunit in complex with different complements of translation initiation factors. Thus, we can now relate much of the structural information to our models of the various substeps of initiation and approach a reasonable reconstruction of the dynamics of the entire process. Nevertheless, some challenges remain for the future. First, we still do not fully understand the mechanism of SSU locomotion during scanning and the structural organization of the scanning “motor”. Second, there is no comprehensive understanding of differential initiation factor usage across transcriptomes in different organismal and cellular contexts. Given the recent progress in cryo-EM and high-throughput sequencing technologies, both problems can now be directly approached, and we can expect some major developments in the coming years. Advances of single-molecule approaches (Halstead et al., 2015; Johnson et al., 2017; Marchetti, Malinowska, Heller, & Wuite, 2017; Morisaki et al., 2016; Wu, Eliscovich, Yoon, & Singer, 2016) and their merger with high-throughput sequencing techniques (Kim et al., 2011; Manos, Camunas-Soler, Croquette, & Ritort, 2017) can be anticipated to result in the deeper insights there.

Another important limitation that needs to be acknowledged is that such a patchwork of insights gleaned from disparate data sources may only approximate the sequence of events for a generic or “average” mRNA. From cumulative evidence over many years, we can anticipate a pronounced and context-dependent variability across mRNAs even within the canonical initiation paradigm. For example, initiation may play out differently depending on the stage of the mRNA lifecycle. Clearly, the first round of translation initiation on just arrived mRNA will have unique features, especially with regard to the initial steps of cap attachment and scanning (Maquat et al., 2010). The importance and extent of the cap-to-poly(A) closed-loop may vary between mRNAs (Archer et al., 2015), even between mRNAs derived from the same gene, as differential promoter usage and alternative 3′ end cleavage and polyadenylation frequently generate isoforms with varying 5′ and 3′ UTRs (Harrison et al., 2015; Turner et al., 2017; Clancy et al., 2011). Aspects of the initiation mechanism may also be executed differently depending on subcellular localization. For example, compared to “free” cytosolic polysomes, ER-attached ribosomes must contend with additional constraints for the formation of closed-loop interactions and transfer of the components during reinitiation. High-throughput sequencing-based methods such as ribosome profiling and TCP-seq will likely provide the transcriptome-wide data to illustrate and dissect these events.

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CONFLICT OF INTEREST

The authors have declared no conflicts of interest for this article.

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